Neural Computing and Applications (2025) 37:63–90
https://doi.org/10.1007/s00521-024-10384-y

ORIGINAL ARTICLE

Groupers and moray eels (GME) optimization: a nature-inspired metaheuristic algorithm for solving complex engineering problems

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Received: 25 November 2023 / Accepted: 24 August 2024 / Published online: 16 November 2024

Abstract

As engineering technology advances and the number of complex engineering problems increases, there is a growing need to expand the abundance of swarm intelligence algorithms and enhance their performance. It is crucial to develop, assess, and hybridize new powerful algorithms that can be used to deal with optimization issues in different fields. This paper proposes a novel nature-inspired algorithm, namely the Groupers and Moray Eels (GME) optimization algorithm, for solving various optimization problems. GME mimics the associative hunting between groupers and moray eels. Many species, including chimpanzees and lions, have shown cooperation during hunting. Cooperative hunting among animals of different species, which is called associative hunting, is extremely rare. Groupers and moray eels have complementary hunting approaches. Cooperation is thus mutually beneficial because it increases the likelihood of both species successfully capturing prey. The two predators have complementary hunting methods when they work together, and an associated hunt creates a multi-predator attack that is difficult to evade. This example of hunting differs from that of groups of animals of the same species due to the high level of coordination among the two species. GME consists of four phases: primary search, pair association, encircling or extended search, and attacking and catching. The behavior characteristics are mathematically represented to allow for an adequate balance between GME exploitation and exploration. Experimental results indicate that the GME outperforms competing algorithms in terms of accuracy, execution time, convergence rate, and the ability to locate all or the majority of local or global optima.

Keywords Optimization · Associative hunting · Nature-inspired · Metaheuristic · Grouper and Moray Eel algorithm

1 Introduction

Optimization refers to the process of identifying the most optimal or superior option among all available solutions. Optimization aims to select input values within a given set of permitted values in order to minimize or maximize one or more defined objective functions. In optimization algorithms, the procedure for locating the optimal solution begins with the random generation of a certain number of solvable solutions that satisfy the problem's constraints. Afterward, these randomized solutions are improved through the utilization of the algorithm's phases, and a procedure relied on repetition. When the algorithm is entirely executed, the best proposed solution for the optimization problem is determined [1].

Despite the fact that many optimization algorithms have been developed, some of them may be highly effective at addressing a particular set of optimization challenges, while being incapable of addressing another set. This is due to the fact that the mathematical models and characteristics of different real-world problems vary. Consequently, the efficiency of handling all optimization issues cannot be guaranteed by any single optimization technique [2]. Real-world optimization problems have become more difficult, necessitating more efficient solution methods, so various

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researchers have studied various methods for dealing with these complex and difficult real-world problems [3]. Certain researchers employ conventional techniques, including conjugate gradient, quasi-Newton, and sequential quadratic programming, in order to address these optimization problems. However, due to the nonlinearity, presence of several decision variables, and complex restrictions in most real-world optimization problems, solving them efficiently becomes challenging using traditional techniques [3]. The advantages of metaheuristic algorithms include that they don't rely on the problem model, don't require gradient information, have a robust search capability, are widely applicable, and can achieve a satisfactory balance between optimal solutions and computational cost. Because of their ability to efficiently generate optimal solutions using computational resources, metaheuristic algorithms have frequently been embraced as an appropriate method to tackle optimization challenges. [4].

To identify the optimal solution, each of these metaheuristic algorithms depends on exploration and exploitation in the search space. Exploration denotes the algorithm's pursuit of potential regions within a vast search space, while exploitation signifies the algorithm's attempt to identify the optimal solution within those potential regions; thus, the solution quality is determined by the equilibrium between the two search behaviors. Maintaining a delicate equilibrium between exploration and exploitation is of utmost importance in metaheuristic algorithms to prevent fluctuations in the convergence rate and to identify both local and global optimal solutions. In order to achieve optimal convergence, metaheuristic algorithms frequently use parameter controls to effectively balance between them [5].

A novel nature-inspired algorithm, namely Groupers and Moray Eels (GME) Optimization, is proposed in this paper to be more effective for solving various optimization problems. GME mimics the associative hunting between groupers and moray eels. GME consists of four phases: primary search, pair association, encircling or extended search, and attacking and catching. The optimization rule in GME typically involves selecting the best solutions and the process continues for a predetermined number of iterations or until a termination criterion is satisfied, such as the desired level of solution quality or the completion of a predetermined number of iterations. This means that as the iterations progress, the population is updated, and the quality of the solutions improves. GME aims to converge towards an optimal or near-optimal solution to the given problem. GME also behaves in balance in the exploration and exploitation phases with an adaptive mechanism over search space. Moreover, the key contributions of this paper are summarized below:

- A new nature-inspired optimization algorithm called GME is proposed to mimic the associative hunting strategy between different species which are: grouper and moray eels, and it is very rare to observe this cooperation in nature.
- Based on the good results obtained in the experimental section of the paper;
    - o The suggested GME algorithm can successfully escape the local optima trap. This happened because it combines evidence from the behavior of two different animals, each of which has a unique movement pattern. This results in good search domain coverage and balanced behavior between the exploitation and exploration phases.
    - o GME is relatively easy, and simple to implement when compared with most metaheuristic algorithms.
- The performance evaluation of the GME algorithm is evaluated by comparing its results to those of other well-known metaheuristic methods. The experimental results suggest that the GME algorithm performs better in solving difficult optimization issues.

The overall structure of this paper is as follows: literature review is provided in Sect. 2. Section 3 describes the inspiration for GME and constructs the corresponding mathematical model. A discussion of the results has been provided in Sect. 4. The conclusion and future research take place in the last section.

2 Literature review

This section is divided into three subsections. An explanation of nature-inspired optimization algorithms is described in Sect. 2.1. On the other hand, cooperative versus associative hunting is explained in Sect. 2.2. Section 2.3 discusses the characteristics of Groupers and Moray Eels Pair.

2.1 Nature-inspired algorithms

In the real world, optimization entails not just enhancing effectiveness but also minimizing time and computational cost. In reality, there are a lot of difficult and complex optimization problems that need to be solved. There are two techniques for resolving issues and obtaining optimized solutions: heuristic and metaheuristic algorithms. In metaheuristic algorithms, the best solution is found from the random search space and predefined boundaries, whereas heuristic algorithms are dependent on a specific problem. Both exact and approximate approaches can be used to address Nondeterministic Polynomial time (NP-hard) issues [6, 7]. The exact procedures, which have a high cost and exponential time complexity, ensure the best solution. The heuristic and metaheuristic algorithms belong to the family of approximate algorithms. Although they cannot be guaranteed to discover the exact solution, they may be closer to it as they have a lower complexity and faster execution time. The second group is typically preferred. Examples of heuristic algorithms are such as; Hill climbing (HC), Simulated Annealing (SA) [8], and Best First Search (BFS) [9]. The metaheuristic approaches can offer a useful and practical solution for many NP-hard problems and typically outperform heuristic methods in finding the best solutions for different types of problems in terms of actual execution time.

The metaheuristic approaches are more flexible and attempt to find the best solutions in global search areas with low processing costs in a short amount of execution time with simple implementation. They also avoid falling into local traps. In particular, metaheuristic algorithms have the potential to be more effective in solving complicated issues [10]. Additionally, according to no free-lunch (NFL), there isn't an algorithm that can solve all optimization problems to the best of its ability [2]. Consequently, there is a significant demand for the development of novel metaheuristic algorithms that may be used for a variety of issues. Metaheuristic optimization algorithms have thus gained popularity in recent years in various scientific fields due to their wide range of applications and benefits. Metaheuristic algorithms are divided into two types: single solution-based algorithms and multiple solutions-based algorithms. When the search process starts with a single candidate solution it is called a single solution-based algorithm, but in multiple solution-based algorithms, the optimization is carried out using a population of solutions. Metaheuristic algorithms with a population perform better than those with a single solution [11, 12].

Exploration and exploitation are the two fundamental phases of metaheuristic algorithms, and each algorithm has its strategy for implementing the principles of both phases. The greater the balance between these two phases in the proposed algorithms, the higher the success rate. The randomness and proper coefficients defined in the algorithms are important. It is critical to carefully adjust the relevant coefficient parameters to achieve a suitable balance between these two phases. The exploration algorithms look for new solutions in new areas, this means the exploration entails a global search. This step needs more searching thus the applicable algorithm chooses solutions at random from the search space. The phase of exploitation follows the exploration phase and focuses on using current solutions and modifying them so their fitness improves. The algorithm's ability to escape traps is influenced by the randomness of the search in exploration [6, 13].

Nature-inspired metaheuristic algorithms are high-level heuristics that draw inspiration from a variety of sources, including physical and chemical systems [14, 15], swarm intelligence, biological systems [16], artificial systems, and so on. Swarm intelligence, physics-based, and evolutionary algorithms are the three main categories of metaheuristic algorithms as shown in Fig. 1. The Darwinian Theory of Evolution serves as a source of inspiration for evolutionary algorithms (EAs) [17, 18]. EAs use a population-based approach for problem-solving, where the entire population has an impact on the best solution. They address problems inside a stochastic search domain and utilize the most optimal solution from the previous iteration to generate new solutions in the following iteration, as the most suitable solutions in the current iteration are more likely to contribute to the creation of a new solution in the next iteration. The genetic algorithm (GA) is a well-known EA algorithm that is inspired by generation reproduction. Another algorithm inspired by nature's evolution is the differential evolution (DE) algorithm. The DE algorithm differs from the GA algorithm in the selection procedure used to generate the next generation. Other examples of an EA are the Biogeography-Based Optimizer (BBO), Tabu Search (TS), Black Window optimization (BWO), and Evolutionary programming (EP) [19]. Physics-based algorithms are another type of metaheuristic technique. These algorithms are inspired by nature's physics rules and operate at random. The search space and optimal solution in this type of algorithm follow physical rules such as gravitational force, electromagnetic force, and mechanical force [20]. This category includes well-known algorithms such as the gravitational local search (GLSA), curved space optimization (CSO), the gravitational search algorithm (GSA), big bang-big crunch (BBBC), and black hole (BH). The third category of metaheuristic algorithms is swarm intelligence (SI). In general, SI approaches are influenced by the social behaviors of creatures that live in nature in swarms, flocks, and herds [16, 21, 22]. In this type of algorithm, search agents attempt to find an ideal solution by incorporating social intelligence. There are many algorithms in this category such as; grey wolf optimization (GWO), particle swarm optimization (PSO), artificial bee colony (ABC), whale optimization algorithm (WOA), and ant colony optimization (ACO). In this paper, many swarm-based algorithms are used to compare their performance with the GME which are; Cheetah Optimizer (CO) [23], Fennec Fox Optimization (FFA) [24], Hermit Crab Optimization Algorithm (HCOA) [25], Improved Salp Swarm Algorithm (ISSA) [26], Osprey Optimization Algorithm (OOA) [27], Red-Tailed Hawk algorithm (RTH) [28], Walrus Optimization Algorithm (WaOA) [7].

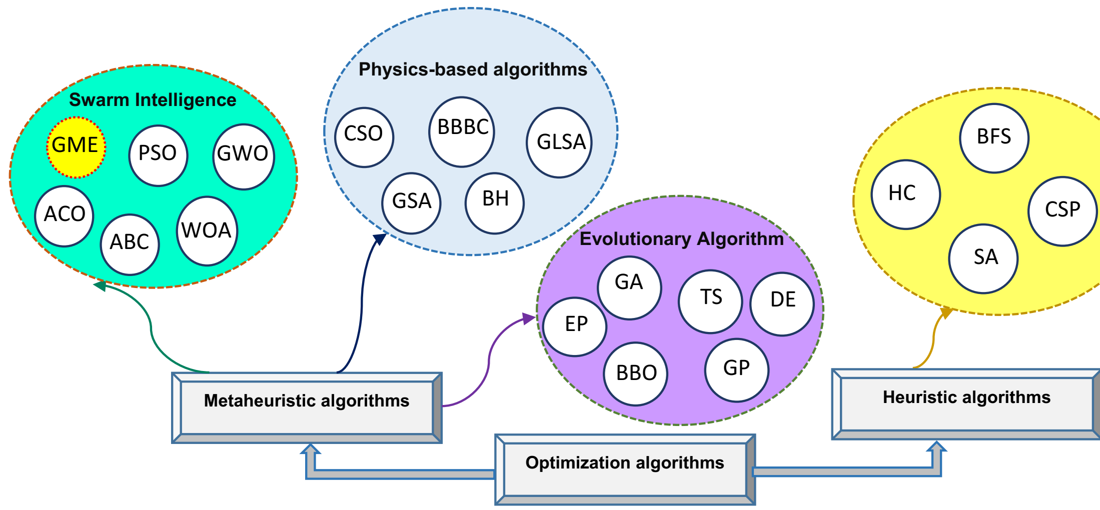


Fig. 1 Different types of optimization algorithms

2.2 Cooperative versus associative hunting

The pursuit of prey may require the collaborative efforts of multiple individuals in order to achieve a successful hunt. When hunting alone, the probability of successfully capturing wildlife that is particularly resistant or nimble may be low. Some have hypothesized that a comparable difficulty has resulted in intricate collaborative hunting conduct among chimpanzees, wherein the participating individuals assume distinct hunting roles. Collaborative hunting can also be beneficial when the prey is too sizable to be subdued by a solitary hunter.

Both cooperative and associative hunting are fired due to one or more factors such as; (i) the prey's energy to continue the chase. For example, the prey may be very fast and have a high ability to continue chasing for a long time, which requires cooperation between the individuals to follow one after the other until the prey loses its ability to continue chasing; (ii) the prey can be slow; and however, prey size plays an effective role in the evolution of cooperative hunting. The large-sized prey certainly needs many individuals to be taken down. For example, a group of lions must cooperate to land an African buffalo [29].

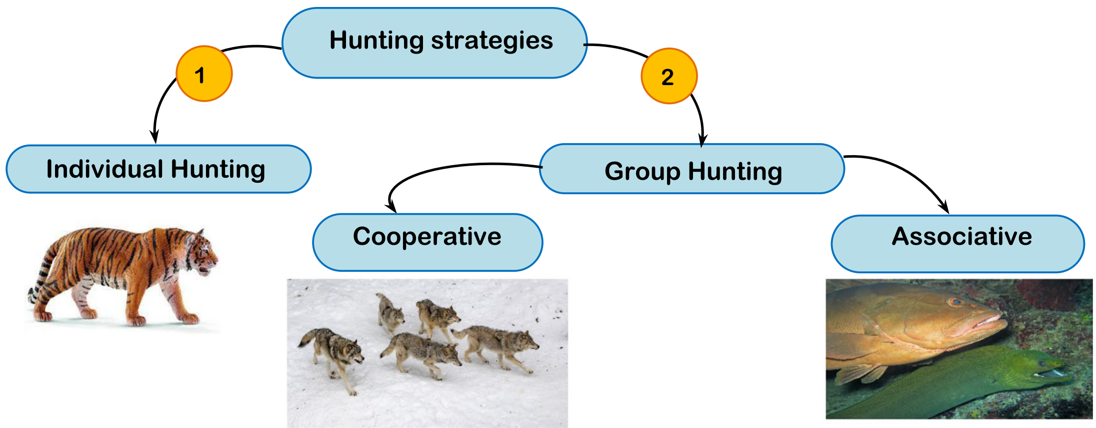
Generally, group hunting maximizes the success rate of catching the prey. Moreover, based on several studies, the larger the group size, (i) the more captures per hunt, (ii) the greater the range of prey that can be taken, and (iii) the greater the likelihood of making multiple kills per hunt [29]. Figure 2 illustrates the difference between cooperative and associative hunting.

2.3 Groupers and moray eels pair

The moray eel and the grouper fish form an unexpected friendship in the vibrant depths of the ocean. Watch as these two species use their unique hunting abilities to conquer the food in the water. A grouper reaches an eel's hiding spot and jerks its head quickly, indicating that it wants to hunt. The eel detects the signal and follows the grouper. The grouper shakes its head again as it guides the eel to a location where prey is located. The grouper cannot penetrate this area, but the eel may enter tight fissures and hunt the prey out. Although either the grouper or the eel will catch a certain prey animal, researchers have discovered that each animal consumes the prey at a different period. As a result, the pair should go hunting together. Moray eels, unlike many predators, have limited eyesight and must rely on their extraordinary sense of smell to hunt. Because moray eels rely primarily on smell, their preferred food is frequently weakened or dead to make it easier to detect. This makes them excellent reef cleaners [30]. The collaboration appeared to benefit both morays and groups, with each species experiencing greater hunting success than when hunting alone, as shown in Fig. 3.

3 Materials and methods

The Theorem of No Free Lunch (NFL) states that there is no assurance that an algorithm that performs well at optimizing a specific group of problems will perform well at optimizing all other optimization problems. We claim that this problem is due to the fact that all previous algorithms assume one model of movement because they depend on agents from the same species and therefore the movement of all of them is constant. NFL pushes researchers to create new metaheuristic algorithms to tackle optimization issues in several fields more successfully. The NFL theorem also influenced the authors of this work to implement a new metaheuristic algorithm depending on agents from



	Individual	Group	
		Cooperative (Social)	Associative
Description	It is a method of hunting in which a single hunter does not enjoy any support, whether from individuals of the same taxa or of another taxa.	Cooperative hunting between the same related taxa, in which every individual works for the benefit of the group.	Cooperative hunting involves taxa that are distantly related, wherein one species facilitates the access of other participants to prey. However, each pursues his own advantage.(i.e. the associative hunting is a cooperative hunting between different species)
Number of agents	Single	Multiple	Multiple
Success rate	Low	Medium	High
Agent type	Single	Single	Multiple
Examples	• Tiger	▪ Lions ▪ Wolves ▪ Whales. ▪ Hyenas.	• Groupers and Moray Eels. • Coyotes and Badgers.

Fig. 2 Different hunting strategies

different species to propose effective optimum results to optimization challenges. In nature, there are many observations about the existence of cooperation between animals of different species such as; Honey Badger and Wolf, and grouper and moray eel. In this research, we will introduce a new algorithm based on cooperation between groupers and moray eels where each of them has a different strategy in the attack. When these two predators combine their hunting abilities, more than one agent is looking for prey with different strategies, and fast access to prey, and prey has little chance of surviving. If the prey hides below the reef, the sinuous shape and ambush techniques of the moray eel come into play. If the victim runs into open water, the grouper's speed and agility will seal its doom. This cooperative behavior appears to be an indication of high intelligence. The cooperation between the groupers and moray eels (i.e. cooperation in hunting between different species) provides better results than the hunting between the same species.

3.1 The proposed groupers and moray eels (GME) optimization

In GME optimization, the flock consists of an even number of search agents. Hence, it can be easily divided later into pairs. Each pair consists of two individuals, the first consists of grouper fishes, and the other one contains the eels as shown in Fig. 4. The pair works together to catch the prey so that each individual has a specific function. There is no overlap in the tasks within the same pair, but the pair individuals work together in a distinct harmony, which increases the possibility of successful prey catching.

The cooperation between grouper fish and Moray eel during the hunting process goes through four phases, which are; (i) Primary search (PS) for a prey, (ii) Pair association (PA), (iii) Encircling or extended search (ES), and (iv) Attacking and catching (AC). GME begins with a population of potential solutions. These solutions are created stochastically within the problem's stated upper (UB) and lower (LB) bounds. Each potential solution to the optimization issue is represented by a candidate solution. The goal is to discover the optimum solution for a certain goal or fitness function. Based on the problem's requirements, the evaluated values for the fitness function are used as the primary criterion for assessing the quality of the proposed solutions. As a result, the best value for the fitness function corresponds to the best candidate solution (i.e., the best search agent), while the worst value corresponds to the worst candidate solution (i.e., the worst search agent) in each iteration of the optimization process. This means that the population is updated as iterations go, and the quality of the solutions improves. In GME, the optimization

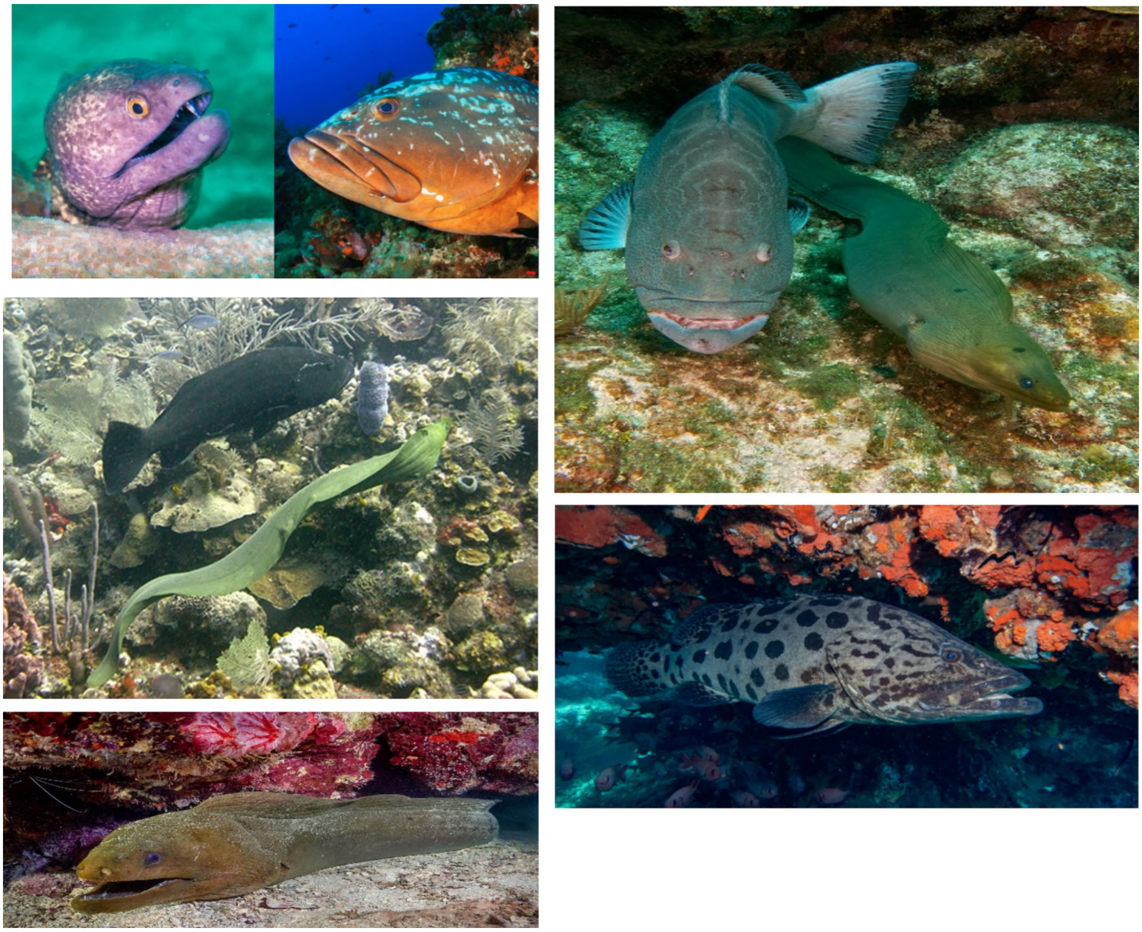
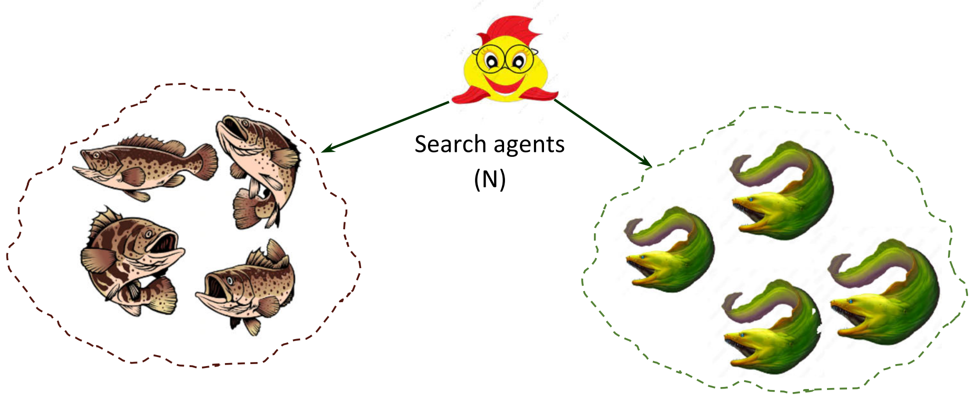


Fig. 3 Cooperation between grouper and moray eels

Fig. 4 Search agent divided into two groups



procedure often entails selecting the best solutions and continuing the procedure for a pre-established number of iterations or until a termination condition is met, such as reaching a maximum number of iterations or achieving a certain level of solution quality. GME tries to converge towards an optimal or near-optimal solution to the given problem by iteratively updating the population and considering the best solution acquired thus far.

During the first phase, which is PS, only the groupers search for the potential prey. Once the PS phase has finished, it is assumed that each grouper fish has located its prey and the task now is to choose one of the eels to pursue and chase that prey among the rocks. This task is accomplished during the PA phase. After pair formulation, each grouper fish has become associated with an eel. Then, each pair tries again to search for prey after it has been hidden among the coral reefs. This is accomplished through the ES phase. However, each agent searches for prey in its own way. The eel searches for prey among the rocks inside the coral reefs, while the grouper spins outside the coral reefs in spiral rings, waiting for the prey to exit from inside the coral reefs, hoping that the eel will not be able to catch it. The final phase is the AC phase, in which the prey is caught by one of the pair of individuals, either the grouper fish or the moray eel. Through the next subsections, each of the pre-mentioned phases will be explained in more detail.

3.2 Primary search (PS) phase

In the wild, groupers are intriguing fish to observe. They can travel through the water by propelling themselves ahead with their bodies. They can also fast shift the direction of their fins. This adaptability allows grouper to move quickly through the water, while avoiding predators and prey. Exploration is mainly carried out during the search phase in all nature-inspired optimization algorithms, whereas encircling and attacking are reserved for exploitation. In order to achieve this objective, the search agents are permitted to navigate the search space at random. The search agents' movements are considered to be random due to the uncertain location of the optimal solution (the prey). Without a doubt, a successful prey search improves the algorithm's exploration ability.

Groupers move in various ways, depending on their habitat and hunting strategy. They use the zig-zag swimming method to navigate their way around. This method allows them to quickly cover a large area and track potential threats. In this phase, the grouper fish is looking for the presence of prey, and its movement is in the form of a zig-zag, where a random place is imposed for each grouper fish, and then each of them begins to move. Figure 5 shows the different forms of zig-zag movement of a grouper fish in the search space.

A zigzag motion occurs when no three successive positions of the movement are in either increasing or decreasing order. In other words, if the array of movement has three elements (h_i, h_{i+1}, h_{i+2}) such that; $h_i < h_{i+1} < h_{i+2}$ or $h_i > h_{i+1} > h_{i+2}$, the movement is not a zig-zag. In brief, the array's elements must be ordered in less than ($<$) and greater than ($>$) order. Every element is bigger or smaller than its neighbors. The arrays X = [1, 2, 4–6, 10] and Y = [8, 3, 6, 2, 5, 0, 8, 9] are zigzag arrays because no three successive array elements are in rising or decreasing order. Under the assumption that P represents the overall number of iterations in the PS, ES, and AC. As for the number of iterations of the PA phase (P_{Ass}), it is equal to 1.

Therefore, the respective values of the number of iterations for PS, ES, and AC denoted as; $\lfloor P_{\mathrm{Search}} \rfloor$, P_{Enc}, and P_{AC} can be calculated using the Eqs. (1–3) as:

$$P_{\mathrm{Search}} = \left\lfloor \frac{P}{3} \right\rfloor \tag{1}$$

$$P_{\mathrm{Enc}} = \left\lfloor \frac{p}{3} \right\rfloor \tag{2}$$

while

$$P_{AC} = \left\lfloor P - 2 * \frac{P}{3} \right\rfloor \tag{3}$$

For more illustration, assuming P = 12, then $P_{\mathrm{Search}} = \lfloor \frac{12}{3} \rfloor = 4$, $P_{\mathrm{Enc}} = \lfloor \frac{12}{3} = 4 \rfloor$, and $P_{AC} = \lfloor 12 - 2*\frac{12}{3} = 4 \rfloor$. Let S be the set of N available search agents (groupers and moray eels), at the beginning, the search agents are divided into two equal sets of groupers and eels. The No. of agents in each set is

$$\mathrm{n} = \mathrm{N}/2 \tag{4}$$

(4). The set of groupers is called G = {g_1, g_2, g_3,………., g_n}, while The set of eels is called E = {e_1,e_2,e_3,……….,e_n}.The next step is to randomly distribute the groupers and eels in the search space using (5). Equation 5 distributes the search agents by generating a random number between 0 and 1, multiplying it by the difference

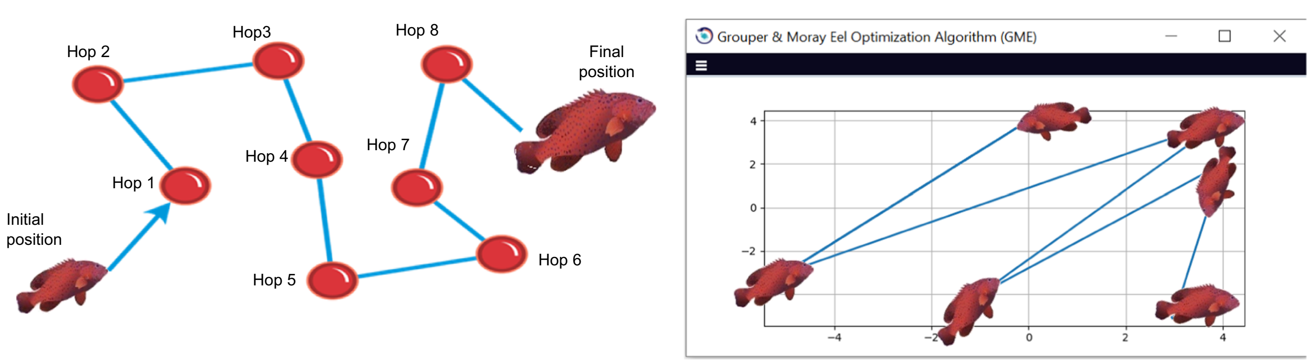


Fig. 5 Searching for the prey by a grouper fish with zig-zag movement

between the upper and lower bounds, and then adding up the lower bound. This ensures that each agent's positions fall within the specified lower and upper boundaries of the search space, as stated in (5) [24].

$$X_{ij}^{\text{initial}} = \text{low}_j + \text{rand}.\left(\text{upper}_j - \text{low}_j\right), i = 1, 2, 3, \ldots\ldots, N, j = 1, 2, 3\ldots.., D \tag{5}$$

where X_{ij}^{initial} is the initial position of ith search agent of jth dimension, *upperandlower* are the upper and lower boundaries of the search space, N is the number of search agents, D is the total number of dimensions, and rand represents a random vector that follows a uniform distribution, with values ranging from 0 to 1.

After that, the objective function is calculated for each grouper. The groupers begin to move in a zigzag form to find the prey (optimal solution). The zigzag movement allows the groupers to discover new regions in search space and increases the exploration ability of the algorithm. At the end of this phase, the best position for each grouper is calculated according to the objective function.

Let, $\overrightarrow{X_{gmj}}^{i} = \left\{X_{gm1}^{i}, X_{gm2}^{i}, X_{gm3}^{i}, X_{gm4}^{i}, \ldots\ldots\ldots\ldots\ldots, X_{gmD}^{i}\right\}$ is the position vector of the mth grouper in the ith iteration, where $1 \leq i \leq p_{search}, 1 \leq j \leq \text{D}, 1 \leq \text{m} \leq \text{n}$, n is the total number of groupers, and p_{search} is the total number of search iterations. The location vector of each grouper is updated after each hop in the current iteration during the PS phase and the corresponding objective function is evaluated, which indicates the proximity of the grouper to the possible prey. At the end of the current iteration, the positions from all hops that have the best value of the objective function will be the initial positions of the next iteration. The updated position of a grouper can be calculated using (6), and it depends on the number of *hop* which is the number of movement in the iteration. If the number of hop is even, then the updated position will be a random position that is greater than the current position but doesn't exceed the search space's maximum boundary; if the number is odd, the updated position will be a random position that is less than the current position but greater than or equal to the search space's minimum boundary.

The optimal position of each grouper in a given iteration is determined by the position vector that yields the maximum value of the objective function when compared to positions created across all the hops of the iteration. After finishing the searching phase, the best position of each grouper is determined as the position vector that yields the highest value of the objective function compared to all other positions of a grouper generated during all iterations of the PS phase. Once the best position is determined, they initiate the PA phase from their respective best positions. As an illustrative example, assume that we have $N = 12$ search agents (groupers and moray eels) in two-dimensional space x_1 and x_2. It is assumed that x_1 and $x_2 \in [-6, 6]$, which is set to be the domain of the search space. The number of groupers = 6, and the number of eels = 6. For simplicity of the illustrative example, we assume that $P_{\text{Search}} = 1, P_{\text{Ass}} = 1, P_{\text{Enc}} = 1, P_{\text{Att}} = 1$ and hop = 3. The initial position vectors for the groupers across search agents are assigned randomly using (5). The employed objective function is $f(X) = x_1^2 - x_1x_2 + x_2^2 + 2x_1 + 4x_2 + 3$ Following that, the grouper fish searches for prey, and its movement is in the shape of a zig-zag in which after a random location is imposed for each grouper fish, each of them begins to move. The steps of the PS phase are shown in Algorithm 1. Consider Table 1: which summarizes the three hops in the first iteration of groupers during the primary search process. At the initial iteration, the position of six groupers is initialized randomly in the search space from [−6,6], and then the grouper begins to move in zigzag form according to no.of hop in the iteration. After the end of the current iteration, the best position from all hops in the iteration is selected and will be the initial position of the next iteration until reached to the total no.of iterations in the PS phase. Table 2 shows the best position of each grouper from the PS phase. Table 3 shows the random positions of the moray eels and the values of the objective function.

$$\text{The updated position of grouper}\begin{cases} X_{gmj}^{\text{hop}+1} = \text{Rand}(X) \quad \text{where}, X_{gmj}^{\text{hop}} < X \leq \max\left(X_{gmj}\right) & \text{if No.of hop is even} \\ X_{gmj}^{\text{hop}+1} = \text{Rand}(X) \quad \text{where}, \min\left(X_{gmj}\right) \leq X < X_{gmj}^{\text{hop}} & \text{if No.of hop is odd} \end{cases} \tag{6}$$

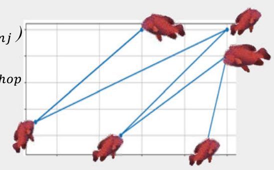
Algorithm 1: The steps of the PS phase

Inputs:
- S: The set of *N* available search agents (Groupers and moray eel).
- F(X): Objective function to minimize or maximize.
- P_{search} : Total number of search iterations.
- hop: Number of movement in the iteration.

Outputs:
- The best positions of the groupers after the searching phase.
- The best objective function of each grouper.

Steps:

1. *Dividing the search agents into two equally groups of grouper and eels; i.e.* $No.of\ agents\ in\ each\ group\ = N/2$
2. *Randomly distribute the groupers across the search space.*
3: *Calculating objective function value for each grouper.*
4: *Each grouper begins to move in the form of zig-zag movement.*
5. *Calculate new position for each grouper based on number of movement in the iteration &calculate new objective function*
6. *For i=1 to* P_{search}
7. *For m=1 to* $\frac{N}{2}$
8. *For k=1 to hop*
9. *If k mod 2==0 then* *// The number of movement is even*
10. $X_{gmj}{}^{hop+1} = Rand(X)$ *, Where* $X_{gmj}{}^{hop} < X \le max(X_{gmj})$
11. *Else* *// The number of movement is odd*
12. $X_{gmj}{}^{hop+1} = Rand(X)$ *, Where* $min\left(X_{gmj}\right) \le X < X_{gmj}{}^{hop}$
13. *End if*
14. *K=k+1*
15. *Next*
16. *Select the best position of the grouper from all hops in the current iteration to be the initial position of next iteration*
17. *Next*
18. *Next*
19. *// assigning the best position for each grouper*
20. *For each grouper* g_m *in G*
21. $Obj_{Val_{max}}(g_m) = Objective\_Value(g_{mj}, 1))$
22: $\vec{X}_{g_{mj}} = \vec{X}_{g_{mj}}{}^{1}$
23: *For i=2 To* $p_{search} - 1$
24: *If* $(Objective\_Value(g_{mj}, i) > Obj\_Val_{max}(g_{mj}))$
25: $Obj\_Val_{max}(g_{mj}) = Objective\_Value(g_{mj}, i))$
26: $\vec{X}_{g_{mj}} = \vec{X}_{g_{mj}}{}^{i}$
27: *End if*
28: *Next*
29: *Next*
30. *Descending order the groupers according to the maximum objective function.*
31: *Randomly distribute the moray eels in the search space and calculate the objective function of each eel.*



As shown in Table 2, the best positions of g1, g3, g5, g6 are obtained from the hop 3. The best value of the objective function of g2 is obtained from the second hop, while the best value of g4 is obtained from the first hop, this is because the optimal position of each grouper in a current iteration is identified as the vector of position that produces the highest value of the objective function when compared to positions generated during all the hops of iteration. Table 3 shows the positions of eels at the start of the PA phase. At the end of this phase, the best position for each grouper is the position that achieves the best value of objective function during all iterations of the PS phase and it will be the initial position of a grouper in the PA phase.

3.3 Pair association (PA) phase

Groupers are adept predators, renowned for their rapid bursts of velocity that render them fearsome adversaries. Nevertheless, their substantial dimensions and unwieldy form prevent them from capturing animals that conceal themselves in narrow crevices and fissures. In such cases, they employ a unique strategy of seeking out moray eels to flush out their elusive prey. Both the grouper and moray eel have distinct predatory abilities. Groupers are predatory fish that employ an ambush strategy, utilizing their large jaws to engulf their prey in its entirety. In addition, they possess the ability to swiftly propel themselves through the water in quick bursts. In contrast, moray eels possess a body structure that enables them to maneuver through narrow openings in the coral reef in order to reach their prey. They lack pectoral fins and gill plates, which eliminates the risk of getting trapped. Through collaboration,

Table 1 The three hops of the groupers during PS phase

Initially				First hop				Second hop				Third hop			
Agent	× 1	× 2	Objective function	Agent	× 1	× 2	Objective function	Agent	× 1	× 2	Objective function	Agent	× 1	× 2	Objective function
g1	2.693	1.235	18.777619	g1	4	1.302	28.6952	g1	2.73	1.289	19.2115	g1	5	1.308	38.4029
g2	2.72	4.23	39.1457	g2	1.62	4.653	41.5889	g2	2.3	4.584	41.6959	g2	1.709	4.591	40.9339
g3	4.233	−0.283	29.5303	g3	5.843	0.84	47.9841	g3	5.73	0.657	46.5879	g3	5.904	0.734	48.8064
g4	3.398	−1.578	22.8825	g4	4.453	−2.047	36.8527	g4	4.298	−1.987	34.6091	g4	4.449	2.456	36.6208
g5	−0.568	3.235	27.4293	g5	−0.738	2.875	23.9560	g5	−0.618	3.025	25.2660	g5	−1.73	3.019	28.9461
g6	2.677	3.608	33.3114	g6	3.608	2.873	32.6140	g6	3.095	3.763	36.3347	g6	4.95	2.23	40.2569

Table 2 The best positions of the groupers from the first iteration

Agent	Best position		Objective function
	× 1	× 2	
g1	5	1.308	38.4029
g2	2.3	4.584	41.6959
g3	5.9	0.734	48.8064
g4	4.45	−2.047	36.8527
g5	−1.7	3.019	28.9461
g6	4.95	2.23	40.2569

Table 3 The locations and objective functions of the eels after random distribution

Agent	× 1	× 2	Objective function
E1	−0.568	1.833	13.91966
E2	3.871	0.0921	25.747
E3	5.113	4.067	51.3827
E4	−3.12	−4.044	−5.94494
E5	−0.874	−0.188	1.134908
E6	−4.213	−4.189	−4.53317

these two carnivores may effectively patrol a larger area for hunting. The grouper operates in the upper water column, while the moray eel navigates the depths of the reef. The utilization of these hunting methods greatly diminishes the likelihood of their target being able to evade capture. When trapped, the prey has two options: either remain concealed until the moray eel retrieves it or take the chance of swimming in the water, where the grouper's formidable teeth lie in wait.

During the PS phase, the search agents are distributed randomly within the search domain, and the corresponding objective function values are calculated and recorded. After the PS, the PA phase occurs, where cooperation between the groupers and the Eels allows them to discover new areas of the search domain. The intelligence of the grouper and its high ability to learn shows that it can choose the best eel in the hunting process. Each grouper fish will have an association with an eel, and this process is called pair identification. There are several mechanisms for dividing the search space into pairs such as; (i) the random association between eels and groupers i.e. the grouper can associate with any eel, (ii)The association based on the nearest distance between eel and grouper, but this requires many calculations, and (iii) the association based on the objective function of groupers and eels i.e. the grouper with the highest value of fitness function associates with the eel that has the highest value of fitness function and so on. The

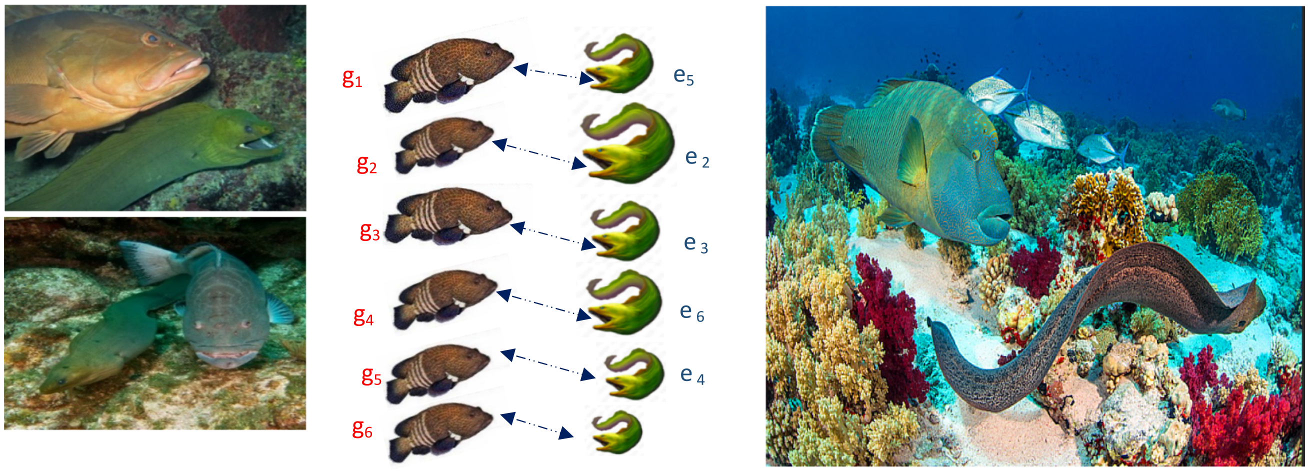
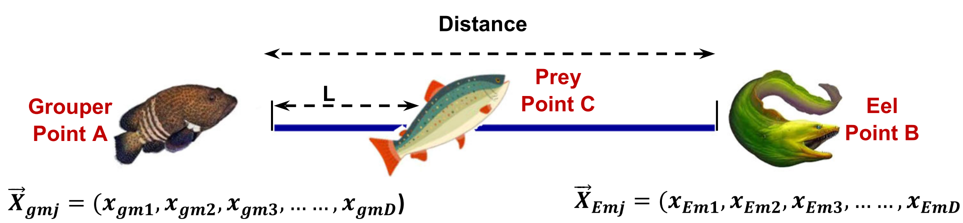


Fig. 6 The association between groupers and eels

Table 4 The association between groupers and eels according to the objective function

Eels in descending order according to objective function				Groupers in descending order according to objective function				Pairs
Agent	× 1	× 2	Objective function	Agent	× 1	× 2	Objective function	
E3	5.113	4.067	51.3827	g3	5.904	0.734	48.8064	(g3, E3)
E2	3.871	0.0921	25.747	g2	2.3	4.584	41.6959	(g2, E2)
E1	−0.568	1.833	13.91966	g6	4.95	2.23	40.2569	(g6, E1)
E5	−0.874	−0.188	1.134908	g1	5	1.308	38.4029	(g1, E5)
E6	−4.213	−4.189	−4.53317	g4	4.453	−2.047	36.8527	(g4, E6)
E4	−3.12	−4.044	−5.94494	g5	−1.73	3.019	28.9461	(g5, E4)

Fig. 7 Locating the position of the prey between a grouper and an eel



last method is applied in this paper. Figure 6 shows the association between groupers and eels according to the objective function. After the PA is finished, the ES phase follows where each search agent moves towards its best position found during the PS phase. In the AC phase, the search agents converge towards the optimal solution. Table 4 illustrates the descending order of eels and groupers according to their fitness function. Pair 1 consists of grouper 3 and eel 3 (g3, e3) which have the best value of objective function, Pair 2 consists of (g2, e2), Pair 3 consists of (g6, E1), Pair 4 consists of (g1, e5), Pair 5 consists of (g4, e6), and Pair 6 consists of (g5, e4) which have the lowest value of objective function.

3.4 Encircling or extended search (ES) phase

Each Pair tries to encircle the prey during this phase, moving independently so they can explore new parts of the search region. In the open ocean, groupers have been seen performing a sort of underwater shimmying dance to call moray eels and indicate that they want to hunt in groups. The “Grouper to Eel Encircling Signal” (GES) is the name of this signal. On rare occasions, they will even perform an underwater headstand, point their heads in the direction of the fish’s hiding place, and then wiggle their bodies to show the neighboring eel, where the prey is lurking. Rarely when an eel ignores the signal, groupers aggressively approach the moray and try to force it towards the intended prey. The moray reacts to the signal by chasing the fish out

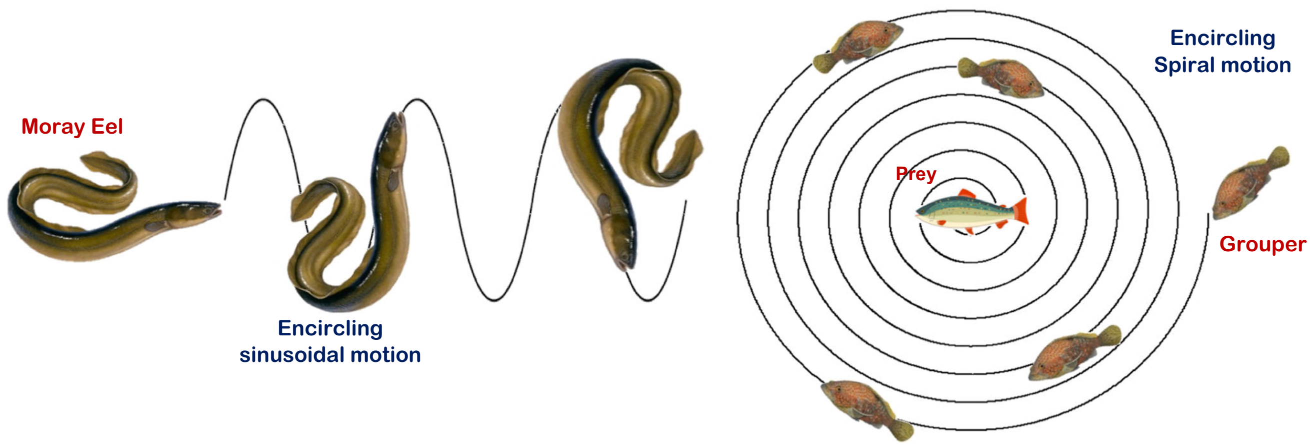


Fig. 8 The grouper and the eel encircle their prey during ES phase

of its hiding place so that it may be caught in the snappy Grouper's jaws. Then they engage in an actual attack on the prey and partake in the meal together, reaping the benefits of their associative hunting techniques.

The position vector of the mth grouper in D dimensional space can be represented as $\vec{X}_{gmj} = (x_{gm1}, x_{gm2}, x_{gm3}, \ldots\ldots, x_{gmD})$, and the position vector of the mth eel in D dimensional space can be represented as: $\vec{X}_{Emj} = (x_{Em1}, x_{Em2}, x_{Em3}, \ldots\ldots, x_{EmD})$. Given the unknown location of the prey, which symbolizes the optimal solution, it is postulated that the prospective prey is situated in the region bounded by the grouper and the eel, as shown in Fig. 7.

In an n-dimensional space, the formula for determining the position of a point that lies between two other points, while maintaining a specified distance, is an extension of the formula used in two dimensions. Let's assume two points, point A (location of grouper) and point B (location of eel). A point that lies between these two points is point C (location of the prey) as shown in Fig. 7. The position of the prey can be calculated as follows: First, Eq. (7) is used to find the differences between the coordinates (Δx_{mj}) of the position of a grouper(X_{gmj}) and the position of an eel (X_{Emj}) in each dimension [31].

$$\Delta x_{mj} = \left(X_{Emj} - X_{gmj}\right) \tag{7}$$

Next, the distance between the grouper and the eel is calculated by squaring the difference for each axis, then summing them up and taking the square root using (8) [31].

Distance Between the grouper and the eel

$$= \sqrt{\sum_{j=1}^{D} \left(\Delta x_{mj}\right)^2} \tag{8}$$

Finally, the coordinates of the prey in each dimension can be calculated using (9) [31].

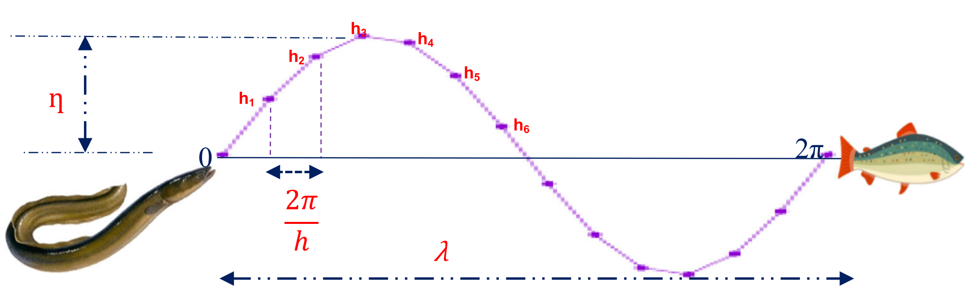
$$c_{mj} = X_{gmj} + \frac{L}{dis}\left(X_{Emj} - X_{gmj}\right) \tag{9}$$

where c_{mj} is the coordinates of the mth prey in each dimension, $x_{gmjD}$is the position of a grouper, x_{EmjD} is the position of an eel, L is the distance between the grouper and prey, and dis is the distance between the grouper and the eel as calculated using (8).

After locating the prey, both the grouper and the eel begin to move toward the prey, each in its own way, as shown in Fig. 8.

The logarithmic spiral has been selected as the primary location update method for groupers during the ES phase. The following requirements must be met for any other types of spiral to be used: (i) the spiral's initial point must be the location of the grouper, (ii) its final point must be the location of the predicted prey, and (iii) the spiral's

Fig. 9 Modelling the eel motion in an iteration during ES phase



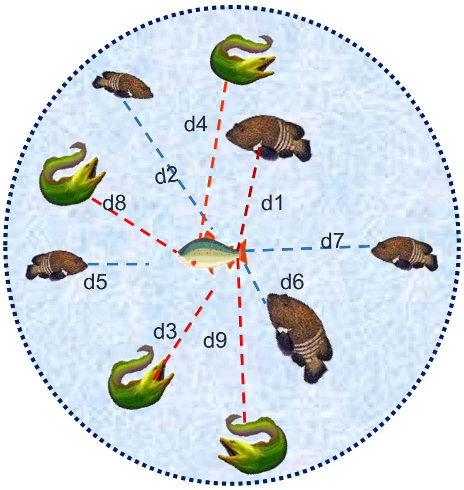


Fig. 10 The first attack circle

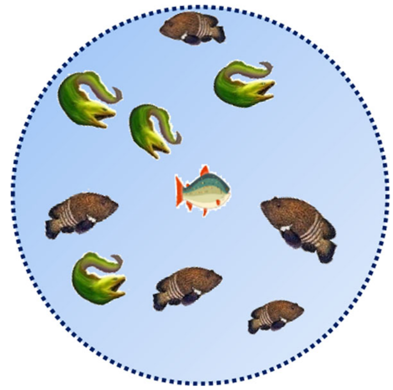


Fig. 11 The second circle

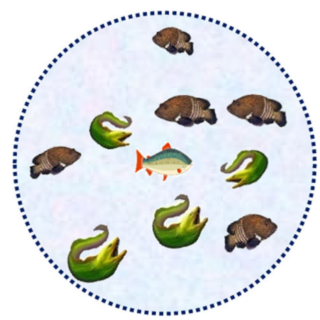
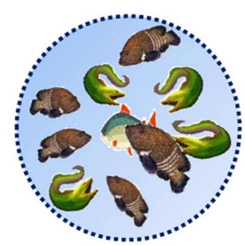


Fig. 12 The third circle

Fig. 13 The nth circle



oscillation range must not exceed the search boundaries. To update each grouper's position, we first calculate the distance between it and its potential prey, which is the difference between their locations using (10).

$$\overrightarrow{D1} = \left|\vec{X}_{prey\_m}(i) - \vec{X}_{gm}(i)\right| \tag{10}$$

The position of a grouper is updated many times in the same iteration according to the number of hops (h). The position with the best fitness function value from all hops after the end of the current iteration will be the initial position of the second iteration and so on, until reached to the determined number of iterations in the ES phase. The spiral equation, as shown in (11) [32], describes the movement of the groupers toward the prey. The updated position of mth grouper in the following hop $\vec{X}_{g_m}(h+1)$ depends on; $\overrightarrow{\mathrm{D1}}$, which is the distance between the mth grouper and the prey, constant k that determines the shape properties of the logarithmic spiral, the predicted location of the mth prey at iteration i $\overrightarrow{(X}_{\mathrm{prey\_m}}(i))$, and w which is a value can be calculated using (12). It depends on $\mathrm{P}_{\mathrm{encircle,}}$ which is the number of encircling iterations, and h is the number of hops in the iteration.

$$\vec{X}_{g_m}(h+1) = \overrightarrow{D1}.e^{kw}\cos(2\pi w) + \vec{X}_{\mathrm{prey\_}m}(i) \tag{11}$$

$$w = 1 - \frac{2*h}{P_{\mathrm{encircle}}} \tag{12}$$

A wave can have many different shapes, sine waves being just one of them. Sine waves are a significant shape because they are frequently found in nature, are simple to handle mathematically, and can be used to create any arbitrary wave shape. As a result, sine waves have been selected as the primary position update mechanism for moray eels during the ES phase, as shown in Fig. 9.

The eel starts to move in a sinusoidal wave, where its position is updated several times in the same iteration according to the number of hops (h). First, the wavelength (i.e. the distance between the eel and the prey $\left(\vec{\lambda}(i)\right)$ will be computed using (13). $\vec{\lambda}(i)$ is the difference between the position of mth eel at iteration i $\vec{X}_{Em}(i)$ and the position of mth prey at iteration i $\left(\vec{X}_{\mathrm{prey\_}m}(i)\right)$.

$$\vec{\lambda}(i) = \left|\vec{X}_{Em}(i) - \vec{X}_{\mathrm{prey}_m}(i)\right| \tag{13}$$

Then, using (14), the wave amplitude $\left(\vec{\eta}(i)\right)$ will be determined by multiplying $\vec{\lambda}(i)$ by a factor ξ ξ has a random value between 0 and 1.

$$\vec{\eta}(i) = \vec{\lambda}(i) * \xi \tag{14}$$

The distance between the eel and the prey is divided into hops. The distance between the current and the next hop is calculated using (15).

$$\text{The distance between the hops} = \frac{2\pi}{\text{Total no. of hops}} \tag{15}$$

The positions of the eels will be updated using (16) [33]. The position of the eel (X_E^{i+1}) depends on α, which is a random value, $\vec{\lambda}(i)$ ξ the value of the sin angle, and the position of the eel in the current iteration.

$$X_E^{i+1} = \alpha * \vec{\lambda}(i) * \xi * \sin(g) + X_E^i \tag{16}$$

After the end of the current iteration, the best position of the eel from all hops of the iteration will be selected to be the initial position of the next iteration.

As an illustrative example for pair 2, which consists of g2 and e2, each iteration has three hops. In each hop, the search agents update their positions according to the form of their movement. In the first iteration, the best position for g2 is obtained from hop 3, while the best of e2 is from hop 2, these positions are used to be the initial positions of iteration 2 as shown in Table 5. The best positions for g2 and e2 are determined in the second iteration at hop 3 and hop 2, respectively; these positions are utilized as the starting points for iteration 3. The best positions in the ES phase are obtained from iteration 3 at hop 3 for g2 and e2. These positions are the initial positions of the next phase. Algorithm 2 shows the steps of modeling the movement of the groupers during the ES phase, while the steps of updating the positions of eels are illustrated in Algorithm 3.

Algorithm 2: Modeling the grouper motion

Inputs:

- G: the set of ($n = N/2$) available search agents of groupers
- E: the set of ($n = N/2$) available search agents of eels.
- $P_{encircle}$: Total number of encircling iterations.
- The best position vector for each jth grouper from the searching phase
- h: number of hops for the grouper's movement.
- k : is a constant defines the shape of the logarithmic spiral

Outputs:

- The best positions of the groupers after the encircling phase.
- The best objective function of each grouper.

Steps:

1. *For i=1 to* $P_{encircle}$
2. *For m=1 to n*
3. Calculating the location of the mth prey using (7, 8,9)
4. *Calculating* the distance between the grouper and the prey using (10)
5. *For q=1 to h*
6. *// Calculate new position for each grouper*
7. *w=1-(2*h)/$P_{encircle}$*
8. $\vec{X}_{g_m}(h+1) = \overrightarrow{D1}.\, e^{kw}\, cos(2\pi w) + \vec{X}_{prey_m}(i)$
9. *Calculate the objective function for the position of the hop.*
10. *q=q+1*
11. *Next*
12. *Select the best position of the grouper from all hops in the current iteration to be the best position for this iteration and the initial position of next iteration.*
13. *Next*

14,Next

15. *// assigning the best position for each grouper*
16. *For each grouper* g_m *in G*
17. $Obj\_Val_{max}(g_m) = Objective\_Value(g_m, 1))$
18. $\vec{X}_{gm} = \vec{X}_{g_m}{}^{1}$
19. *For i=2 To* $P_{encircle}$
20. *If* $(Objective\_Value(g_m, i) > Obj\_Val_{max}(g_m))$
21. $Obj\_Val_{max}(g_m) = Objective\_Value(g_m, i))$
22. $\vec{X}_{g_m} = \vec{X}_{gm}{}^{i}$
23. *End if*
24. *Next*
25. *Next*

Algorithm 3: Modeling the eel motion

Inputs:

- E: the set of (n) available search agents of moray eels.
- $P_{encircle}$: Total number of encircling iterations.
- h : Number of hops in the iteration.

Outputs:

- The best positions of the eels after the encircling phase.
- The best objective function for each eel.

Steps:

1. Calculate the wave length

$$\vec{\lambda}(i) = \left|\vec{X}_{Em}(i) - \vec{X}_{prey_m}(i)\right|$$

2. *Calculate* the wave amplitude

$$\vec{\eta}(i) = \vec{\lambda}\,(i) * \xi$$

3. Dividing the distance between the eel and prey according to *number of hops where the distance between each hop* $= \frac{2\pi}{h}$.

4. Calculate new position for each eel

5. *For i=1 to* $P_{encircle}$

6. *For m=1 to* $\frac{N}{2}$

7. *For* $g = \frac{2\pi}{h}$ *to* $= 2\pi - \frac{2\pi}{h}$ *step* $\frac{2\pi}{h}$

8. $X_E{}^{i+1} = \alpha * \vec{\lambda}(i) * \xi * \sin(g) + X_E{}^{i}$

9. *Calculate the objective function for each position*

10. *Next*

11. *Select the best position which has the high value of objective function to be the initial position of the next iteration .*

12. *Next*

13. *Next*

14. *// assigning the best position for each eel*

15. *for each eel* E_m *in E*

16. $Obj\_Val_{max}(E_m) = Objective\_Value(E_m, 1))$

17. $\vec{X}_{Em} = \vec{X}_{E_m}{}^{1}$

18. *For i=2 To* $P_{encircle}$

19. *If* $(Objective\_Value(E_m, i) > Obj\_Val_{max}(E_m))$

20. $Obj\_Val_{max}(E_m) = Objective\_Value(E_m, i))$

21. $\vec{X}_{E_m} = X_{Em}{}^{i}$

22. *End if*

23. *Next*

24. *Next*

3.5 Attacking and catching phase

At the AC phase, all search agents, whether groupers or eels, participate in the attack on the prey after accurately surrounding its location. In 2D space, the hypothesis relies on forming a circle around the prey, where the prey is located in the center of the circle. Figures 10, 11, 12, 13 show the nth attack circles, where the search agents in each circle are assumed to be nine, as the number of attacking circles (AB) is the number of attack iterations. In 3D space, the hypothesis relies on forming a ball around the prey called the "attack ball" as shown in Fig. 14.

The steps for forming the circle around the predicted prey are as follows; at first, the location of the agent with the best fitness function is assumed to be the location of the predicted prey. Then the distance between the predicted prey and all other agents is calculated. After that, a circle around the prey is formed and the radius (R) of the circle is the distance between the prey and the farthest agent. For the large number of search agents and to reduce the complexity of computation, the radius of the next circle can be calculated using (17).

$$R_{i+1} = (1 - \mu) * R_i \tag{17}$$

, where i = 1, 2, 3, ….., AB-1 and μ is a shrinking ratio. For example, if the distance between the prey and the farthest agent is equal to 100 and μ = 0.2, then the radius of the first circle (R_1 = 100). The radius of the second circle is equal to (1−0.2)*100 = 80. The radius of the third is equal to (1−0.2)*80 = 64 and so on.

After determining the radius of the second circle, the search agent will be randomly distributed within the circle by using the steps of Algorithm 4. In each attacking iteration, the search agents get much closer and closer to the prey as shown in Figs. 10, 11, 12, 13. The process works until the number of attacking iterations is reached, and finally, the agents catch the prey in the last iteration. Algorithm 5 shows the steps for generating random points inside a sphere. Algorithm 6 shows the steps of generating random points inside the N-dimensional sphere**.** Algorithm 7 shows the steps of the AC phase.

Algorithm 4: Generating random points inside a circle

// Steps for generating random points inside the circle
1. Set the radius of the circle and create a random angle that ranges from 0 to 2π, signifying a random point inside the circle in polar coordinates.
2. Use the following formulas to transform these polar coordinates to Cartesian coordinates:
3. X = R_i* cos (angle)
4. Y= R_i* sin (angle).
5. Repeat the previous steps to produce random points equal to the number of search agents inside the circle.

Algorithm 5: Generating random points inside a sphere

// Steps for generating random points inside the sphere
1. Produce random values for theta (θ) and phi (ϕ) which represent the spherical coordinates of a point in the sphere where the value of θ can range from 0 to 2π and the value of ϕ may fluctuate anywhere from 0 to π.
2. Apply the following formulas in order to transform these spherical coordinates to Cartesian coordinates
3. X = R_i * sin(φ) * cos(θ)
4. Y = R_i* sin(φ) * sin(θ)
5. Z = R_i * cos(φ)
6. Repeat the previous steps in order to generate number of search agents at random inside the sphere.

Algorithm 6: Generating random points inside N-dimensional sphere

// Algorithm for generating random points inside N –dimensional sphere
1. Set the number of dimensions and radius of the N-dimensional sphere.
2. Generate random coordinates in N-dimensional space with a Gaussian distribution (normal distribution); these random coordinates are random points in N-dimensional space.
3. Determine the Euclidean length norm of these coordinates.
4. Rescale these coordinates to have a radius length. This step assures that the points are distributed uniformly across the N-dimensional sphere with the specified radius.
5. Repeat the procedure until the desired number of random points has been generated.

Algorithm 7: The steps of AC phase

Inputs:

- S: the set of available search agents.
- P_{AC} : Total number of attacking iterations.
- AB: No.of attack balls where AB=P_{AC}
- The best positions of groupers and eels from the encircling phase.
- The shrinking ration (μ)=0.2

Outputs:

- The location of the predicted prey.

Steps:

// the steps for forming the first circle around the predicted prey

1. Determine the location of search agent that has the best fitness function to be the location of the predicted prey.
2. Calculate the distance between the predicted prey and all other agents.
3. Calculate the distance between the prey and the farthest agent and set to be the radius (R) of the circle where the prey is located in the center of this circle.
4. Randomly distribute the agents inside the circle using algorithm 4.
5. Calculate the fitness function for each search agent.
6. Select the search agent which has the best value to be the prey in the next iteration.

// The updated positions of the search agent

7. For j=1 to AB-1
8. The radius of the next circles will be calculated as follows;

$$R_{j+1} = (1 - \mu) * R_j$$

9. After determine the radius, the circle is formed and prey will be located in the center of it.
10. Randomly distribute the search agent inside it .
11. Calculate the fitness function for the search agent.
12. Select the search agent which has the best value of fitness function will be the prey
13. Next

3.6 Case study: feature selection using GME optimization algorithm

Due to the exponential rise in data, the quality of information (data) is essential for gradually merging data processing through pattern recognition, data mining, image processing, and various ML techniques. Curse of dimensionality (CoD) is one of the NP-hard challenges in data science because of the big-scale data sets. Therefore, several researchers have applied nature-inspired algorithms to solve large-scale data analytics difficulties. Due to overfitting in the ML model, large-dimensional data may result in the majority of noisy, irrelevant, and redundant data. Dimensionality reduction techniques might be used to address these problems, and the use of machine learning (ML) algorithms should be done as a part of the preprocessing stage. The two most popular dimensionality reduction techniques are feature selection (FS) and feature extraction (FE), where FS cleans up the noisy, redundant, and inconsequential data, potentially improving the

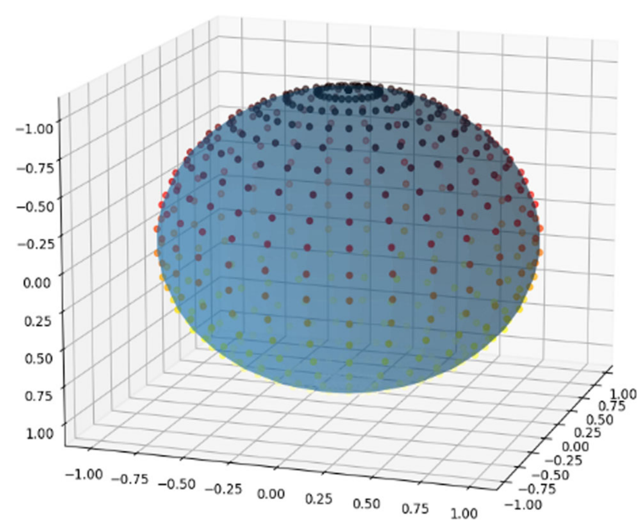


Fig. 14 The search agent form the attacking ball around the prey in the search space

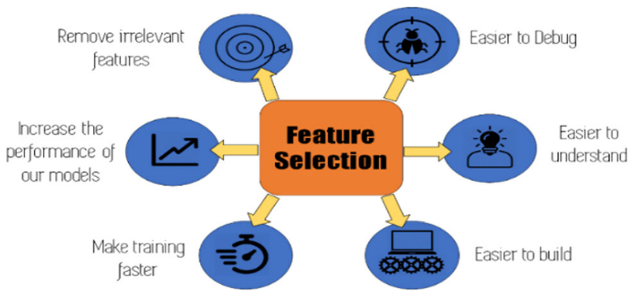


Fig. 15 The advantages of feature selection

effectiveness of an algorithm [34, 35]. Choosing the best features will reduce computational modelling costs while increasing the accuracy of ML models. Furthermore, it increases learning convergence while decreasing storage demand. As a result, metaheuristic algorithms are now being employed to overcome these problems [36]. Figure 15 depicts the benefits of sectioning the best features.

GME is designed to handle continuous optimization problems in which each agent changes its position inside a real-valued search region bounded by the constraints of the particular task. FS is a binary optimization issue in which the search agents can only use binary 0 and 1 values. As a result, without adjustments, the algorithm described above cannot be employed to solve these issues. It is necessary to create a binary version of GME called (BGME) that can be applied to the FS issue. Each solution in this study is denoted by a one-dimensional vector, with its length determined by the number of features in the dataset. Each element in the vector can be assigned one of two values: 1 or 0. A value of 1 indicates that the corresponding feature is selected, while a value of zero indicates that the feature is not selected, as shown in Fig. 16. As a result, to employ the GME for the FS problem, a mapping mechanism from actual values to binary space should be applied. The suggested BGME's usefulness will be assessed in FS by identifying the most significant features for diagnosing Coronary Heart Disease (CHD). Using the proposed GME, the most informative features for CHD diagnosis will be chosen during the FS.

The process of applying the proposed BGME to choose the optimal subset of features from the CHD dataset involves several consecutive steps. BGME commences with a search agent (S) possessing a collection of search agents denoted by Y. For instance, S consists of N search agents, hence including $Y = \{Y_1, Y_2, \ldots, Y_m, \ldots, Y_N\}$. Each agent, denoted as Y_m, within the set S indicates a potential solution. In other words, a solution would indicate the

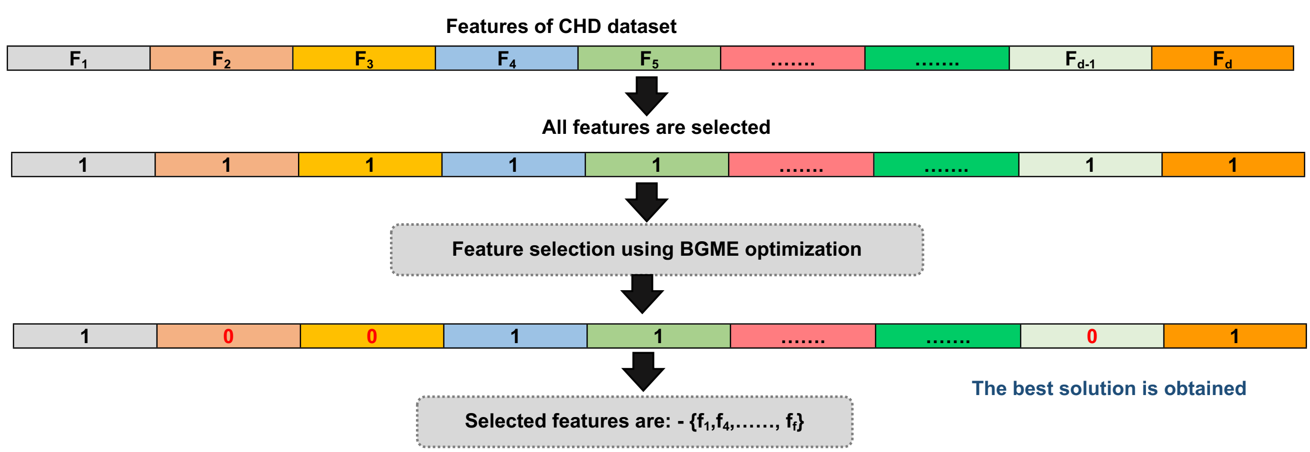
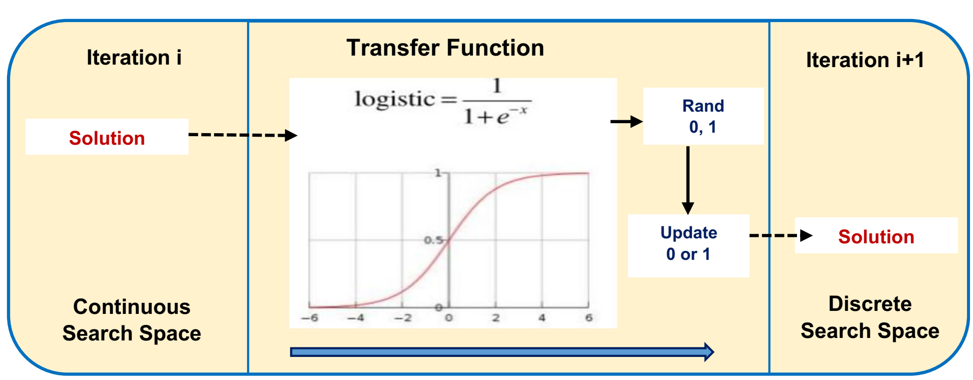


Fig. 16 The representation of the selected/unselected features

Fig. 17 Mapping process from continuous search space to discrete search space



number of features that have been selected to minimize the dimensionality of a particular data set. This process is repeated for every employed population in the F-dimensional search space, where F represents the total number of features. In the CHD dataset, the mth agent can be represented as $Y_m = (Y^1{}_m, Y^2{}_m, \ldots, Y^f{}_m)$.

The position (bit) value (i = 1, 2,…, f) of each agent is a binary value that represents whether the ith feature picked (one) or not chosen (zero). The assessment of these search agents should be conducted by utilizing the accuracy value of the KNN classifier as a fitness function. This evaluation is performed after randomly generating S, which consists of N search agents in a binary space. The evaluation function can be mathematically represented using (18).

$$\text{Fitness}(Y_m) = \text{Acc}\mathit{of}\text{KNN}(Y_m) \tag{18}$$

where the fitness value of the mth search agent is denoted as; Fitness (Y_m), the accuracy of the KNN classifier, denoted as Acc of KNN (Y_m), is calculated based on the subset of selected features in the mth search agent. The collection of groupers and moray eels (search agents) represented by S is distributed randomly in the search space. At the beginning, it is necessary to determine the total number of iterations (P) to calculate the total number of iterations in each phase of GME, and the number of hops for the zig-zag movement (*hop*).In the PS phase, the groupers begin to move and the position of each grouper is updated according to the number of hops if it is even or odd. The mth grouper's new position has a continuous value; as a result, a transfer function must be applied in order to transform this position into a binary value. The transfer function is regarded as a key component of metaheuristic-based FS methods due to its ability to map the continuous search areas into discrete search areas.

The most prevalent example of a sigmoid function that can be applied to translate the continuous values of the positions of the search agents to the binary values is the logistic sigmoid function. It has an "S"-shaped curve when plotted [37], and can be calculated using (19).

$$\text{sigmoid}\left(Y_m^i\right) = \frac{1}{1 + e^{-Y_m^i}} \tag{19}$$

where the natural logarithm's base is e, and the expression "rand (0, 1)" represents a random value that falls within the range of 0–1. Based on a solution's position, it creates a probability value, and then uses this probability value to convert the solution to binary as shown in Fig. 17. The binary solution's component can be updated between 0 and 1 and vice versa. Sigmoid ($Y^i{}_m$) is the sigmoid function that depicts the likelihood that the ith bit will take a value of 0 or 1 as determined by applying (20) [37]. If the value of rand (0,1) is greater than or equal to the value of sigmoid ($Y^i{}_m$), then the binary value of the updated position of the mth individual at the mth bit $\left(Y^i_{binary_m}(i+1)\right)$ will be 1, otherwise, it will be 0.

$$Y^i_{binary\_m}(i+1) = \begin{cases} 1 & \mathit{if\ rand}(0,1) \geq \mathit{sigmoid}\left(Y_m^i\right) \\ 0 & \mathit{otherwise} \end{cases} \tag{20}$$

Based on the updated position $Y^i_{\text{binary}\_m}(i+1)$ for each individual, the fitness function in (18) is used to evaluate each individual in S. During their search for the prey, each individual in S stores the updated positions and the evaluation values. Up until the P_{search} is finished, the PS phase's steps will be carried out. The optimal location for the agent at the end of the PS phase in S is the position that enables it to achieve the highest fitness value during its search for prey. After the objective function is calculated for all groupers and eels, they will be ordered in descending order. The intelligence and high ability to learn of the grouper fish shows that it can choose the best eel in the hunting process. Each grouper fish will have an association with an eel. Theoretically in the associative phase, the best grouper (i.e. the grouper with the best value of fitness function) will associate with the best eel (i.e. the eel with the best value of fitness function), and so on for the rest of search agents i.e., the association will be according to the value of the fitness function.

In the encircling phase, both the grouper and the eel begin to surround the prey, as each of them moves from the best positions of the previous phase. According to the number of encircling iterations (P_{encircle}) and the number of hops, the grouper begins to move in the form of a logarithmic spiral, and the eel moves in the form of a sinusoidal wave. At first, the positions of the potential prey $\vec{X}_{\text{prey}}(t)$will be calculated for each pair, and to transform the continuous values of positions to binary values, the sigmoid function will be used. Then, the groupers and eels will update their positions according to the equations explained in the previous section. The fitness function is used to evaluate the updated positions. The positions that achieve the best values of fitness functions will be used to be the initial positions of the search agents in the AC phase.

The search agent with the highest fitness function value will become the prey. Then, the distance between all other agents and the expected prey is determined. The circle's radius (R) is set to be equal to the distance between the prey and the farthest agent. To accommodate the enormous number of search agents and to simplify computation, the shrinking ratio is used to determine the radius of the following circles using (17). The fitness function will be calculated to locate the predicted prey and draw a circle with a new radius around it. The search agent will be dispersed at random inside the circle once its radius has

Table 5 The results of the illustrative example during the ES phase for pair 2 (g2, e2)

Agent 1 (g2,e2)													
Iteration 1	Grouper position		Eel position		Prey position		Hop number	New grouper position		Objective function	New eel position		Objective function
	X1	X2	X1	X2	X1	X2		X1	X2		X1	X2	
	2.3	4.584	3.871	0.0921	4.185	−0.80628	1	5.677	3.459	52.746	3.851	0.335	25.695
							2	2.325	6	59.106	3.979	0.699	27.291
							3	6.000	4.998	65.984	3.884	0.430	26.092

Agent 1 (g2,e2)													
Iteration 2	Grouper position		Eel position		Prey position		Hop number	New grouper position		Objective function	New eel position		Objective function
	X1	X2	X1	X2	X1	X2		X1	X2		X1	X2	
	6	4.998	3.979	0.699	3.57480	−0.1608	1	5.587	4.921	61.795	4.144	1.0304	29.378
							2	4.009	5.251	54.616	4.377	1.496	32.586
							3	6.000	5.395	69.316	4.359	1.4612	32.333

Agent 1 (g2,e2)													
Iteration 3	Grouper position		Eel position		Prey position		Hop number	New grouper position		Objective function	New eel position		Objective function
	X1	X2	X1	X2	X1	X2		X1	X2		X1	X2	
	6	5.395	4.376	1.496	4.05120	0.71620	1	5.8352	5.7182	70.9236	4.505	1.754	34.497
							2	5.8000	5.8600	72.032	4.583	1.911	35.708
							3	6.00	5.8546	73.5676	4.580	5.904	64.563

been determined. The procedure continues until the number of attacking iterations (P_{AC}) is reached, at which the agents capture the prey in the last iteration. Finally, the fittest search agent provides the best solution with the best combination of features. It should be emphasized that separating iterations into four phases (PS, PA, ES, and AC) allows the BGME algorithm to deliver a quick and accurate subset of features. Also, it allows each phase to try numerous times to find the optimum solution before executing the next phase, and the following phases are implemented more rapidly and precisely. The feature has a "1' value in the fittest solution is a selected feature but if the value is 0, this feature is not selected. Optimizing feature selection can lead to cost reduction in computational modeling and enhance the accuracy of ML models. Moreover, it enhances the rate at which learning reaches a stable state while reducing the amount of storage required.

4 Results and discussion

This section evaluates the GME algorithm's performance and compares it with seven well-known algorithms which are; CO [23], FFA [24], HCOA [25], ISSA [26], OOA [27], RTH [28], WaOA [7]. The experiments were executed using Python on an Intel(R) Core(TM) i7-8550U CPU @ 1.80 GHz 2.00 GHz with 16 GB RAM running Windows 11. To assess the efficacy of the proposed algorithm in comparison with existing optimization algorithms, we conducted a thorough comparison between our proposed BGME algorithm and other optimization algorithms. There are many common parameters for the proposed algorithm as well as the other algorithms. These parameters are such as; Population size, and number of iterations. The population size is set to 25, 50, 75, and 100. The number of iterations is set to 60, 120, 180, and 240. To assess the performance of the proposed algorithm in comparison to all baseline algorithms, we employed three metrics: classification accuracy, no.of selected features, and execution time. The following subsection will discuss one-by-one evaluation metrics results.

4.1 The used dataset and employed parameters

Heart disease is an umbrella term encompassing several illnesses that impact the anatomy and function of the heart. CHD is a cardiovascular condition characterized by insufficient delivery of oxygenated blood to the heart due

Table 6 Features of CHD dataset

Feature	Description
Age	Years of age
Sex	Sex (1 represents males and 0 represents females)
CP	Type of chest pain:
	Value 1: angina typical
	Value 2: Angina atypical
	Value 3: non-angina discomfort
	Value 4: Asymptomatic
Trestbps	stationary blood pressure at the time of hospital admission, measured in millibars Hg
chol	Cholesterol serum concentration in milligrams per deciliter
FBS	Fasting blood glucose levels exceeding 120 mg/dL (1 = true; 0 = false)
RECG	Resting electrocardiographic findings (RECG)
	Value 0: normal
	Value 1: ST-T wave abnormality (T wave inversions and/or ST elevation or depression greater than 0.05 mV)
	Value 2: left ventricular hypertrophy that is probable or certain according to Estes' criteria
thalach	Attainment of the maximal heart rate
Exang	Exercise induced angina (1 = yes; 0 = no)
Old peak	ST depression induced by exercise as opposed to relaxation
Slope	The slope of the ST segment during maximal exercise
	Value 1: upsloping
	Value 2: flat
	Value 3: down sloping
Ca	The quantity of main vessels (0–3) that are coloured with fluoroscopy in order to calcify the vessels
Thal	Outcomes of the nuclear stress test are as follows: three indicate a normal condition, six a fixed defect, and seven a reversible defect
Num	Target variable denoting the angiographic disease status (diagnosis of cardiac disease) in any major vessel
	Negative 50% diameter reduction (Value0)
	Value 1: Diameter narrowing exceeding 50%

to narrowed or blocked arteries. It is the primary cause of death in the United States. According to the Centers for Disease Control and Prevention, there are 18.2 million American adults who suffer from coronary artery disease, which is the predominant form of heart disease in the nation. Therefore, it is crucial to accurately diagnose cardiovascular disease [38]. The CHD Dataset is utilized to do a comparative analysis between the proposed BGME algorithms and the other optimization algorithms [39]. The Cleveland dataset consists of 303 observations, 13 features, and 1 target attribute. Furthermore, the thirteen features encompass supplementary patient data that is pertinent to the non-invasive diagnostic test results described before. The target variable represents the result of the invasive coronary angiography operation, indicating whether or not the patient has coronary artery disease. A score of 0 signifies the lack of CHD, whereas labels 1–4 indicate the existence of CHD. The primary aim of most studies utilizing this dataset has been to distinguish between the presence (values 1, 2, 3, 4) and absence (value) of a particular phenomenon. The dataset's characteristics are explained in Table 6. Figure 18 displays a snapshot from the CHD Dataset.

All nature-inspired algorithms generally require both common control parameters (also referred to as regular parameters) and algorithm-specific control parameters (often referred to as dependent parameters). Commonly employed are control parameters that are not reliant on the specific problem, such as population size, number of dimensions, number of iterations, and so on. The problem-dependent parameters refer to control parameters that are specific to the algorithm being used. The values of these parameters are subject to variation depending on the problem. For illustration, the Genetic Algorithm (GA) necessitates the specification of a mutation probability and a crossover probability, whereas the particle swarm optimization (PSO) demands the determination of an inertia weight and learning factors. These dependent variables may impact the algorithm's performance. As shown in Table 7, the proposed algorithm may seem to use more

A	B	C	D	E	F	G	H	I	J	K	L	M	N
age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	num
63	1	1	145	233	1	2	150	0	2.3	3	0	6	0
67	1	4	160	286	0	2	108	1	1.5	2	3	3	2
67	1	4	120	229	0	2	129	1	2.6	2	2	7	1
37	1	3	130	250	0	0	187	0	3.5	3	0	3	0
41	0	2	130	204	0	2	172	0	1.4	1	0	3	0
56	1	2	120	236	0	0	178	0	0.8	1	0	3	0
62	0	4	140	268	0	2	160	0	3.6	3	2	3	3
57	0	4	120	354	0	0	163	1	0.6	1	0	3	0
63	1	4	130	254	0	2	147	0	1.4	2	1	7	2
53	1	4	140	203	1	2	155	1	3.1	3	0	7	1
57	1	4	140	192	0	0	148	0	0.4	2	0	6	0
56	0	2	140	294	0	2	153	0	1.3	2	0	3	0
56	1	3	130	256	1	2	142	1	0.6	2	1	6	2
44	1	2	120	263	0	0	173	0	0	1	0	7	0
52	1	3	172	199	1	0	162	0	0.5	1	0	7	0
57	1	3	150	168	0	0	174	0	1.6	1	0	3	0
48	1	2	110	229	0	0	168	0	1	3	0	7	1
54	1	4	140	239	0	0	160	0	1.2	1	0	3	0
48	0	3	130	275	0	0	139	0	0.2	1	0	3	0
49	1	2	130	266	0	0	171	0	0.6	1	0	3	0
64	1	1	110	211	0	2	144	1	1.8	2	0	3	0

Fig. 18 Snapshot from the CHD dataset

specific control parameters compared to other algorithms. However, as will be shown in the results, the high efficiency of the GME has been proven using this relatively limited number of parameters when compared with other algorithms that may use a smaller number of specific parameters but give bad results.

4.2 Evaluating the performance of BGME and comparing it with other optimization algorithms

The Figs. 19, 20, 21, 22 show the comparisons between different algorithms based on execution time. The execution time is measured according to different number of iterations i.e. (60, 120, 180, and 240), and different numbers of search agents i.e. (25, 50, 75, and 100). The results demonstrate that the proposed GME has shown better execution time than all other algorithms. At no.of search agent = 25, and no.of iteration = 60, the execution time of CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME is 404, 313, 560, 765, 652, 740, 810 and 218. At no.of search of search agent = 25 and no.of iteration = 240, the execution time of CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME is 1248, 904, 1205, 1390, 1789, 2020, 2056, and 680. The results of execution time at no.of search agent = 50, and no.of iteration = 60 for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME is 780,594,990,1374,1340,1376,1432 and 459. The results of execution time for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME is 1789, 1460, 1973, 2286, 2116, 2639, 2369, and 1043 for no.of iterations = 240. Furthermore, when the number of iterations equals 60, and the no.of search agents equals 75, the execution time is 1298,1 289, 1234, 1560, 1532, 1604, 1757 and 1123, and when the no.of iterations = 240, the results obtained are 2123, 1707, 2119, 2562, 2641, 2903, 2569, and 1584 for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME, respectively. Finally, the results are 1567, 1480, 1643, 1966, 1984, 1918, 1934, 1210 at no.of search agents equals 100 and no.of iterations equals 60 but when the no.of search agents equals 240, the results are; 2314, 2110, 2780, 3080, 2916, 3014, 2943, and 1921 for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME, respectively. Consequently, the accuracy and execution time of the BGME technique increase gradually with the number of iterations and the number of search agents as shown in Figs. 19, 20,21, 22, 23, 24, 25, 26. The smallest value of the execution time is for GME at no.of iteration = 60 and no.of search agents = 25, but the largest value is for ISSA at no.of iteration = 240 and no.of search agents = 100. The best values for the execution time according to different no. of search agents and no.of iterations is for GME. Table 8 summarizes the results of the execution time in seconds for all optimization algorithms.

The accuracy comparisons for various algorithms can be measured in terms of the number of search agents (25, 50, 75, and 100) and the number of iterations (60, 120, 180, and 240) as illustrated in Figs. 23, 24, 25, 26. The proposed GME has demonstrated the largest value of accuracy. As shown in Fig. 23, at an iteration count of 60 and number of search agents equal to 25, the accuracy for CO, FFA,

Table 7 The Specific control parameters for the different algorithms

	Parameter	Description	Assigned Value
GME	N	Number of search agents	25, 50, 75, and 100
	P	Total number of iterations	60, 120, 180, and 240
	hop (h)	Number of hops in the grouper's movement	3
	K	Constant define the shape of logarithmic spiral	Each agent has its own value
	H	Number of hops in the eel's movement	3
	ξ	Factor between 0 and 1	[0,1]
	α	Random value	0.3
	Θ	Angle may fluctuate anywhere from 0 to 2π	0 to 2π
	Φ	Angle may fluctuate anywhere from 0 to π	0 to π
	μ	Shrinking ratio to determine the radius (R)	0.2
CO	$\hat{r}^{-1}_{i,j}$	Randomization term	0.4
	%1.%2 $\alpha^{t}_{i,j}$	%1.%2 Randomization term	%1.%2 $\times$ t/T
	$\check{r}_{i,j}$	Turning factor reflects the sharp turns of the cheetahs in the capturing mode	0.5
	$\beta^{t}_{i,j}$	interaction factor between the cheetah and leader	0.6
	r_1	Random number	Random number from [0, 1]
FFA	R	r is a random number	Random number from [0, 1]
	I	I is a random number	Random number from the set {1, 2}
HCOA	GD(α, δ)	The Gaussian distribution, characterized by a mean value α and a standard deviation δ, is employed to model the distribution of houses	$P_{candidate}{}^{t+1}(\gamma) = GD(\alpha, \delta)$
	GD(β,λ)	The Gaussian distribution, characterized by its mean β and standard deviation, is employed to model the distribution of dwellings	Candiate_pbest^{t+1}(γ, ω) = GD(β, λ)
ISSA	C_1	C_1 is used to balance between exploration and exploitation	$c_1 = 2e^{-(\ 4I\ L)2}$
	C_2	C_2 are random numbers that are uniformly distributed between 0 and 1	Random number [0, 1]
	C_3	C_3 are random numbers that are uniformly distributed between 0 and 1	Random number [0, 1]
	R	r is a random numbers that are uniformly distributed between 0 and 1	Random number [0, 1]
	η_m	ηm is a non-negative number that quantifies the dispersion index	1
	P_m	Pm $\in$ (0, 1) is the probability of the new candidate solution moving according to a probability distribution	0.5
	δ^{j}_{k}	δ j k is a numerical representation derived from the chosen probability distribution	Depend on the probability distribution
	P_a	fraction (P_a) of the elite and humble candidate solutions	0.2
OOA	$r_{i,j}$	Random number	Random number [0, 1]
	$I_{i,j}$	$I_{i,j}$ are random numbers	Random number from the set {1, 2}
RTH	A	A denotes the angel gain and its range between [5–15]	15
	R0	represents the initial value of the radius and its range between [0.5–3]	0.5
	R	r is a control gain and its range between [1, 2]	1.5
WaOA	$I_{i,j}$	$I_{i,j}$ is integer selected randomly between 1 or 2 and it is used to increase the algorithm's exploration ability	2
	$rand_{i,j}$	$rand_{i,j}$ is a random numbers from [0, 1]	Random from the interval [0, 1]

HCOA, ISSA, OOA, RTH, WaoA, and GME is as follows: 66.3, 70.2, 75.66, 74.2, 81.2, 68.9, 78.32, and 79.2. At 240 iterations and 25 search agents, the accuracy of the following algorithms is determined: CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME: 68.53, 74.02, 79.73, 78.04, 83.72, 74.43, 82.34, and 84.12. As the number of

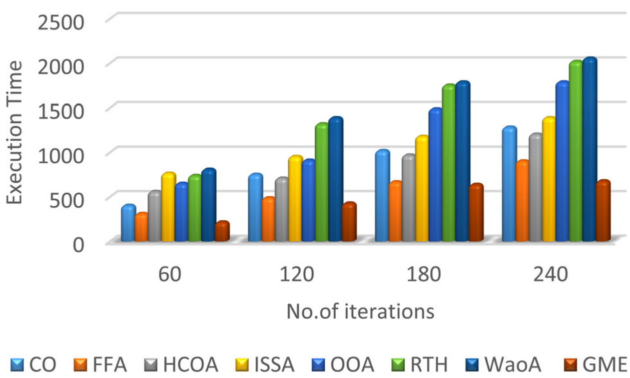


Fig. 19 Execution time of GME and its competitors according to no of search agent = 25

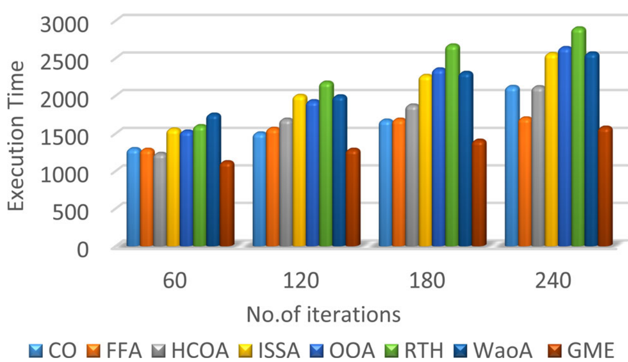


Fig. 20 Execution time of GME and its competitors according to no of search agent = 75

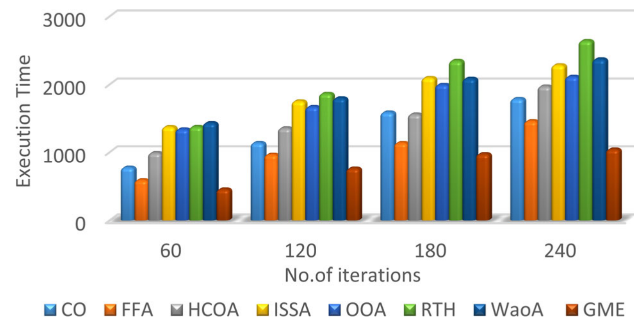


Fig. 21 Execution time of GME and its competitors according to no of search agent = 50

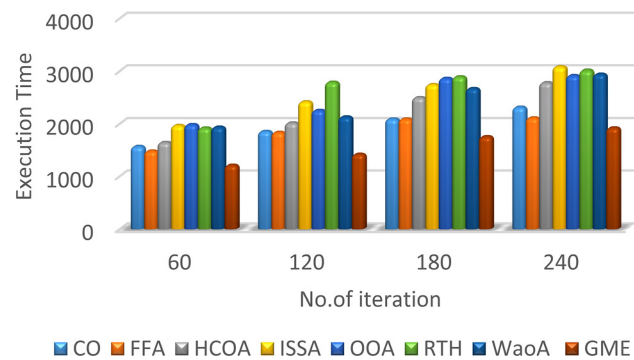


Fig. 22 Execution time of GME and its competitors according to no of search agent = 100

iterations is 60 and the number of search agents is 50, the accuracy for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME is 72.4, 76.11, 80.60, 78.6, 85.09, 76.60, 85.6, and 78.50, respectively. For 240 iterations, the results for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME are as follows: 76.70, 78.30, 82.60, 84.32, 88.35, 84.5, 78.90, and 92.13 as shown in Fig. 24.In addition, when the number of search agents is 75 and the number of iterations is 60, the accuracy is as follows: 79.4, 80.1, 84.77, 86.40, 89.2, 86.05, 89.54, and 94.67. Similarly, when the number of iterations is set to 240, the corresponding accuracy for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME is 82.45, 80.9, 86.5, 88.9, 94.10, 89.21, 93.65, and 97.4 as shown in Fig. 25. When the number of iterations is 60 and the number of search agents is 100, the final results are 83.10, 82.04, 78.3, 89.53, 94.88, 91.3, 94.11, and 97.87. However, when the number of search agents is increased to 240, the results are as follows: 83.25, 84, 88.70, 90.87, 95.70, 92.89, 95.21, and 98.6 for CO, FFA, HCOA, ISSA, OOA, RTH, WaoA, and GME, respectively as illustrated in Fig. 26.

As illustrated in the Figs. 23, 24, 25, 26 the accuracy of the BGME method progressively improves as the number of iterations and search agents increases. The smallest value of accuracy for GME is at iteration count = 60 and number of search agents = 25, while its largest value is at iteration count = 240 and number of search agents = 100. GME provides the most optimal accuracy values when varying the number of search agents and iterations. Table 9 provides a summary of the accuracy of each optimization algorithm. Table 9 demonstrates that the proposed BGME has shown better classification accuracy than all other algorithms on the CHD dataset. BGME outperformed all other algorithms with average accuracy 15.38% higher than CO,14.63% higher than FFA,9.93% higher than HCOA, 7.76% higher than ISSA, 2.93% higher than OOA, 5.74% higher than RTH, and 3.42% higher thanWaoA.

The average amount of selected features for BGME and other optimization algorithms across 240 iterations is presented in Table 10. In comparison to other algorithms,

Table 8 The execution time in (sec) according to different no of iterations and different no of search agent

Optimization algorithm	No.of search agents = 25				No.of search agents = 50			
	Numbers of iterations				Numbers of iterations			
	60	120	180	240	60	120	180	240
CO	404	753	1020	1284	780	1139	1589	1789
FFA	313	489	669	904	594	965	1138	1460
HCOA	560	710	970	1205	990	1356	1560	1973
ISSA	765	954	1180	1390	1374	1753	2098	2286
OOA	652	912	1489	1789	1340	1670	1997	2116
RTH	740	1321	1753	2020	1376	1864	2346	2639
WaoA	810	1390	1789	2056	1432	1798	2087	2369
GME	218	430	640	680	459	767	975	1043
Optimization algorithm	No.of search agents = 75				No.of search agents = 100			
	Numbers of iterations				Numbers of iterations			
	60	120	180	240	60	120	180	240
CO	1298	1507	1678	2123	1567	1854	2090	2314
FFA	1289	1568	1690	1707	1480	1835	2090	2110
HCOA	1234	1690	1879	2119	1643	2015	2498	2780
ISSA	1560	2004	2270	2562	1966	2417	2744	3080
OOA	1532	1932	2356	2641	1984	2254	2864	2916
RTH	1604	2180	2675	2903	1918	2790	2893	3014
WaoA	1757	1997	2310	2569	1934	2130	2670	2943
GME	1123	1289	1410	1584	1210	1419	1754	1921

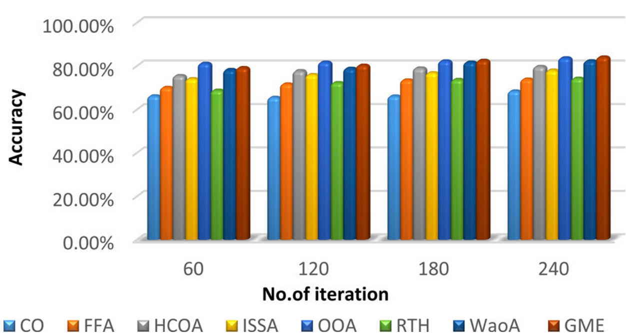


Fig. 23 Accuracy of GME and its competitors according to no of search agents = 25

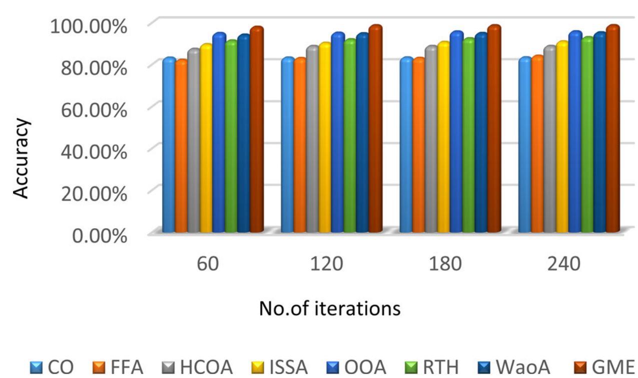


Fig. 24 Accuracy of GME and its competitors according to no of search agents = 50

BGME has chosen the fewest minimum number of features in the CHD Dataset, as illustrated in Table 10. In contrast, the classification accuracy of BGME achieved on the dataset was 98.63. In comparison to all other algorithms, it is evident from this observation that BGME achieves a robust equilibrium between the minimum number of selected features and the classification accuracy. Consequently, this analysis suggests that the BGME prioritizes the selection of informative features that are critical for achieving higher accuracy in classification. Fig. 27 shows the convergence rate of the GME optimizations algorithm and the error goes approximately to zero at 360 iterations.

5 Conclusions and future research

Finding the ideal combination of a group of decision variables to address a certain issue is the process of optimization. Optimization has been used in a wide range of fields, disciplines, and applications. Some problems require

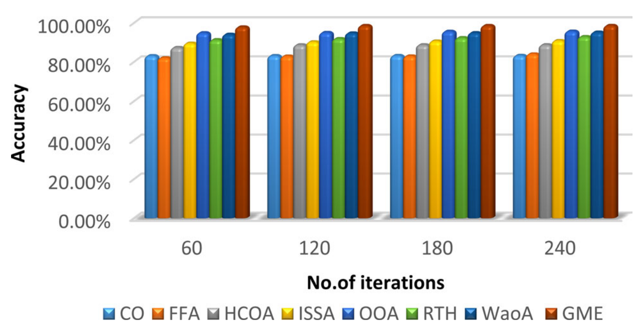


Fig. 25 Accuracy of GME and its competitors according to no of search agents = 75

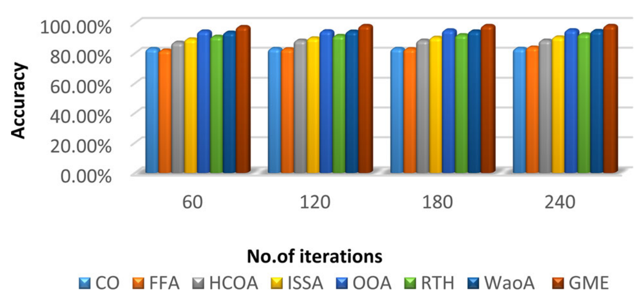


Fig. 26 Accuracy of GME and its competitors according to no of search agents = 100

Table 10 The no of selected features by GME and other optimization algorithms

No.of features in data set = 13	
Optimization algorithms	No.of selected features
CO	11
FFA	10
HCOA	9
ISSA	9
OOA	8
RTH	9
WaoA	8
GME	7

solutions, while others require improvements to their present best solution where the need for more sustainable solutions is continuously growing. To handle these problems, this paper proposed a new swarm intelligence optimization algorithm inspired by the cooperation in hunting between different species (groupers and moray eels) which is extremely rare. While, this associative behavior appears

Table 9 The accuracy of GME and other optimization techniques according to different no of iterations and different no of search agents

Optimization algorithm	No.of search agents = 25				No.of search agents = 50			
	Numbers of iterations				Numbers of iterations			
	60 (%)	120 (%)	180 (%)	240 (%)	60 (%)	120(%)	180 (%)	240 (%)
CO	66.30	65.60	66.21	68.53	72.40	73.02	74.44	76.70
FFA	70.20	71.80	73.65	74.02	76.11	77.40	77.98	78.30
HCOA	75.66	77.80	79.03	79.73	80.60	81.22	81.94	82.60
ISSA	74.20	76.10	76.97	78.04	78.60	79.31	82.05	84.32
OOA	81.20	81.75	82.30	83.72	85.09	86.44	86.90	88.35
RTH	68.90	72.40	73.86	74.43	76.60	78.53	81.30	84.50
WaoA	78.32	78.90	81.74	82.34	85.60	86.19	87.09	87.90
GME	79.20	80.30	82.56	84.12	87.50	89.40	90.89	92.13
Optimization algorithm	No.of search agents = 75				No.of search agents = 100			
	Numbers of iterations				Numbers of iterations			
	60 (%)	120 (%)	180 (%)	240 (%)	60 (%)	120 (%)	180 (%)	240 (%)
CO	79.40	79.63	81.30	82.45	83.10	83.15	83.20	83.25
FFA	80.10	80.34	80.67	80.90	82.04	82.90	82.96	84.00
HCOA	84.77	85.64	86.06	86.50	87.30	88.67	88.68	88.70
ISSA	86.40	87.30	87.65	88.90	89.53	90.20	90.63	90.87
OOA	89.20	91.80	93.50	94.10	94.88	95.02	95.63	95.70
RTH	86.05	88.10	88.92	89.21	91.30	91.87	92.35	92.89
WaoA	89.54	91.42	91.98	93.65	94.11	94.76	94.90	95.21
GME	94.67	96.14	96.88	97.40	97.87	98.59	98.60	98.63

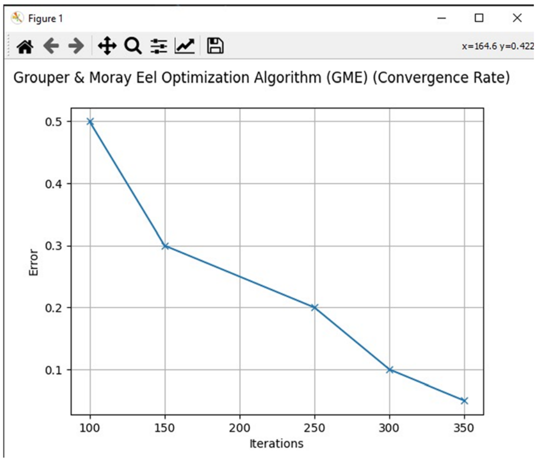


Fig. 27 The convergence rate for GME optimization technique

to be evidence of high intelligence, it could be the product of evolutionary adaptation driven by the rewards of combined hunting. The grouper and moray eel may have simply discovered that hunting together boosts their chances of success over time. The two predators have complementary hunting abilities, and an associated hunt creates a multi-predator attack that is difficult to evade. The marine environment still has so many secrets we have yet to uncover.

GME consists of four phases which are; primary search, pair association, encircling or extended search, and attacking and catching. The balance of exploitation and exploration is a key component of GME;s success. One of GME's advantages is that it requires less time because it requires fewer iterative steps. Using the No Free Lunch theory from optimization, the binary version of GME (BGME) has the potential to outperform existing algorithms in binary issues like feature selection. According to the experimental results, GME outperformed the other algorithms and earned the best overall performance. Also, as a result of its capacity to select the most informative features, while discarding unnecessary data, we recommend this algorithm as an alternative to solving binary problems. Furthermore, the suggested algorithm is capable of investigating unexplored areas by other state-of-the-art algorithms. Future research could focus on the performance of integrating the GME algorithm with other optimization methods or investigating the performance of the GME on a broader range of real-world problems. It is also suggested that the proposed hunting strategies be hybridized with other evolutionary algorithms.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). This research did not receive any funds from any of the supporting institutions or companies.

Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Human or animal rights This paper does not contain any studies with human participants or animals performed by any of the authors.

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